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26 June 1980
SUPERSEDING
MIL-W-52128B
10 March 1969

MILITARY SPECIFICATION

FOR

CRUSHER, JAW, DIESEL-ENGINE AND ELECTRIC-MOTOR

DRIVEN, 75-TON-PER-HOUR CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pneumatic-tired, diesel-engine- and electric-motor-driven, single-pass, jaw-type rock-crusher, semitrailer mounted with dolly converter.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

W-C-586	- Conduit Outlets and Entrance Caps, Electrical; Cast Metal - for Shore Use.
W-R-550	- Rod, Ground with Attachements.
FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.

FSC 3820

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipement Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FF-B-187	- Bearing, Roller, Tapered.
QQ-S-781	- Strapping, Steel, and Seals.
TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
WW-C-566	- Conduit, Metal, Flexible.
WW-C-581	- Conduit, Metal, Rigid; and Coupling, Elbow, and Nipple, Electrical Conduit; Zinc-Coated.
GGG-J-51	- Jacks, Mechanical and Hydraulic, Hand Operated.
KKK-L-370	- Lining, Friction (Clutch and Brake; Metallic, Metal-Ceramic, and Semi-metallic).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape, Packaging, Waterproof.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine (Heavy Duty) .
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control, Semi-Flexible, Flexible and Extra Flexible, 300 and 600 Volts).
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3774	- Crates, Wood; Open, 12,000 and 16,000 Pound Capacity.
MIL-G-3859	- Grease Guns, Hand, High-Pressure, Lever-Operated; Cartridge and Bulk Loading (14-Ounce and 21-Ounce Capacity).

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MIL-H-3992	- Hose and Hose Assembly, Rubber, Air and and Vacuum Brake, Automotive.
MIL-E-10062	- Engines; Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-S-12514	- Starters, Motor; Across-the-Line, and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric.
MIL-C-13486/12	- Cable Special Purpose, Electrical Low Tension, Heavy Duty Multi-Conductor, Unshielded.
MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-A-52363	- Air Cleaners, Intake: Dry-Type (for Internal-Combustion Engines).
MIL-C-52950	- Crates, Wood, Open and Covered.

STANDARDS

MILITARY

MIL-STD-100	- Engineering Drawing Practices.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of.
MIL-STD-423	- Chain, Roller; Power Transmission and Conveyor, Flat Link Plates, Single Base, Single and Multiple Strand, Connecting Links and Attachment Links.
MIL-STD-423	- Chain, Roller; Conveyor, Standard and Large Rollers, Flat Link Plates, Double Pitch, Single Strand; Connecting Links and Attachment Links.
MIL-STD-424	- Chain, Roller; Power Transmission, Offset Sidebars, Single Pitch.
MIL-STD-642	- Identification Marking of Administrative, Combat and Tactical Transport Vehicles.

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MIL-STD-882	- System Program Safety Requirements.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1410	- Method of Selection of Industrial Engines for End Item Application.
MS39127	- Tray Screen Sizes and Edging for Aggregate Screens.
MS51337	- Coupler, Drawbar, Ring - Heavy Duty.
MS53007	- Plate, Identification: Emergency and Service, Automotive Air Brake.
MS53038	- Kingpin, Fifth Wheel - Universal.
MS53039	- Socket, Kingpin, Fifth Wheel - Universal.
MS53040	- Retainer, Kingpin, Fifth Wheel - Universal.
MS53063	- Indicator, Air Cleaner, Intake; Restriction Mechanical Type.
MS75020	- Connector, Plug Electrical l2-Contact, Intervehicular, 24-Volt, Waterproof.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

ICS	- Circuit Breakers, and Controls Molded Case, Industrial.
MG-1	- Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

DEPARTMENT OF TRANSPORT

DOT Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

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(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, 710 North Capitol Street, NW, Washington, DC 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A48 - Gray Iron Castings.
A128 - Specification for Austenitic Manganese-Steel Castings.
A148 - Specification for High Strength Steel Castings for Structural Purposes.
A193 - Alloy-Steel Bolting Materials for High Temperature Service.
A242 - Standard Specification for High Strength Low Alloy Structural Steel.
A536 - Ductile Iron Castings.
C131 - Resistance to Abrasion of Small Size Coarse Aggregate.
C170 - Method of Test for Compressive Strength of Natural Building Stone.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

B29.1 - Precision Power Transmission Roller Chains, Attachments and Sprockets.
B55.1 - Specification for Multiple V-Belt Drives.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanica1 Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

NATIONAL SAND AND GRAVEL ASSOCIATION

Circular No. 80.

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(Application for copies should be addressed to the National Sand and Gravel Association, 527 Munsey Building, Washington, DC 20004.)

CONVEYOR EQUIPMENT MANUFACTURERS ASSOCIATION (CEMA)

Standard No. 502 - Bulk Handling Conveyors, Troughing and Return Idlers.

(Application for copies should be addressed to Conveyor Equipment Manufacturers Association, 1000 Vermont Avenue, NW, Washington, DC 20005.)

RUBBER MANUFACTURERS ASSOCIATION, INC.

Engineering Standard, Specifications for Drives Using Narrow V-Belts (3V, 5V, and 8V Cross Sections).

(Application for copies should be addressed to the Rubber Manufacturers Association, Inc., 444 Madison Avenue, New York, NY 10022.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC.
Automotive Data Book.

(Application for copies should be addressed to the Friction Materials Standards Institute, Inc., 370 Lexington Avenue, New York, NY 10017.)

AMERICAN SOCIETY OF AGRICULTURAL ENGINEERS

ASAE R276.

(Application for copies should be addressed to the American Society of Agricultural Engineers, P. 0. Box 229, St. Joseph, MI 49085.)

FORGING INDUSTRY ASSOCIATION

Forging Industry Handbook.

(Application for copies should be addressed to the Forging Industry Association, Illuminating Building, Cleveland, OH 44101.)

DEPARTMENT OF LABOR (DOL)

Office of Safety and Health Administration (OSHA)

29 CFR 1910 General Industry Standards
29 CFR 1926 Construction Industry Standards

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC.

RP#9 - Lighting Devices for Trailers and Containers.
RP#12 - Trailer Air Brake System Test.
RP#14 - Heavy Duty Lighting System for Truck Trailers.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, 1413 K Street NW, Washington, DC 20005.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street NW, Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

3. REQUIREMENTS

3.1 Description. The portable-jaw crusher shall consist of a pneumatictired semitrailer on which are mounted a diesel-engine-driven jaw-crushing unit, an electric-motor-driven type feeder with material receiving hopper, live grizzly screen, delivery chute or side delivery conveyor, and discharge conveyor. Hoppers, chutes, gates, and all auxiliary fastenings, mountings, guards, operator's platform, controls, drives, and components shall also be mounted on the semitrailer. A dolly converter shall be provided.

3.1.1 Capacity. When crushing quarry rock as specified in 3.1.1.1 which requires 80 percent crushing, the jaw crusher shall have one of the following capacities:

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(a) The crusher shall produce aggregate at the rate of 75 tons per hour (tph) with the jaws adjusted to a closed setting of not more than 2 inches.
(b) When the crusher is used as a primary crusher in conjunction with a secondary 30-inch-diameter by 24-inch-width roll crusher, the capacity shall be compatible for the production of 75 tph of material passing a 1-l/2-inch square screen opening from the secondary crusher.

3.1.1.1 Quarry rock. The quarry rock shall be diabase or traprock having an average compressive strength of not less than 32,000 pounds per square inch (psi) when tested in accordance with ASTM C170, and shall have a percentage of wear not exceeding 20 percent when tested in accordance with ASTM C131.

3.1.2 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance and repair utilizing common hand tools and special tools furnished with the crusher. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage and minimum disturbance to other components of the crusher. Ease of maintenance provisions shall incorporate, to the maximum practicable extent, features insuring operating clearances and facilitating maintenance and servicing operations. Intricate locking devices and controls with threaded fastenings which can be overtorqued by operators shall not be used. Covers or plates which must be removed for component adjustment or repair shall be equipped with heavy-duty quick-disconnect fastenings. Unless otherwise specified herein, covers, plates, panels and guards which must be removed for adjustment, maintenance, or component removal shall be held in place by nonremovable studs, washers, lockwashers, and nuts. Where practicable access shall be provided by a hinged door.

3.2 First article (preproduction model). The contractor shall furnish a crusher for examination and test within the time frame specified to prove, prior to starting production, that his production methods and choice of design detail will produce crushers that comply with the requirements of this specification (see 6.3). Examination and tests shall be those specified herein.

3.3 Initial production. When specified, the contractor shall furnish to the Government one or more crushers for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein (see 6.7). Material not specified shall be selected by the contractor as being compatible with the intended use in accordance with the engineering standards of the earthmoving and construction equipment industry and as set forth in the publications of the American Society for Testing and Materials (ASTM), Society of Automotive Engineers including SAE Handbook (SAE), American Society of Agricultural

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Engineers (ASAE), and American National Standards Institute Inc., (ANSI). Unless otherwise specified herein, rubber products furnished shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the crusher by the Government. Threaded parts shall be in accordance with SAE standards.

3.5 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. The identification of part numbers and changes thereto shall be in accordance with the item identification and part number requirements of MIL-STD-100. Common hardware and fasteners such as bolts, nuts, and screws shall be held to a minimum number of types and sizes. Unless otherwise specified herein, bolts, screws, and studs in sizes 1/4 through 1-1/2 inches shall conform to SAE J429, Grade 5 or stronger, steel nuts in sizes 1/4 through 1-1/2 inches shall conform to SAE J995, Grade 5 or stronger, spring-type lockwashers shall conform to SAE J489, and rivets shall conform to SAE J492. All screw threads shall be in accordance with FED-STD-H28.

3.6 Safety. The safety characteristics of the crusher shall conform to all applicable requirements of the DOT Motor Carrier Safety Regulations, for this type equipment. The manufacturer shall provide evidence or certification to shown that a system safety hazard analysis has been performed on the crusher in accordance with MIL-STD-882. Initial setting-up of the crusher shall not require operating personnel themselves under suspended components or in hazardous positions. For exposed components and systems which are subject to high temperatures, high pressures, electrically or hydraulically actuated, or inherently hazardous, the safety characteristics shall provide safeguarding, enclosing, and insulating features adequate to eliminate the hazard. OSHA (Occupational Safety and Health Administration) regulations shall be followed where applicable. Failure of any hydraulic, electrical or mechanical system shall not produce a condition that could cause personal injury or unreasonable equipment damage. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Step surfaces shall provide a nonskid footing. Nonfunctional sharp edges, projecting points, and excessive length of fasteners shall be eliminated. An unobstructed space shall be provided on the rear of the crusher for mounting of an emblem identifying slow-moving vehicles. A mounting bracket socket shall be permanently installed and shall conform to SAE J725 and ASAE R276. An emblem and its mounting blade shall be furnished on each crusher and shall be in accordance with ASAE R276. For other than travel, the emblem and blade shall be stowed in the toolbox.

3.7 Hoppers and chutes. All portions of hoppers (except the hopper on the feeder), chutes, plates, and metal surfaces subjected to abrasive wear caused by aggregate movement shall be lined or faced with replaceable steel plates conforming to ASTM A242 not less than 3/8 inch in thickness. All bolt heads subjected to the flow of aggregate shall not extend beyond surfaces exposed to the material and shall be of hardened steel.

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3.8 Power units.

3.8.1 Diesel engine. For the purpose of selecting the engine, the crusher shall be classified as a Class I end item in accordance with MIL-STD-1410. The diesel engine including all systems and accessories shall conform to MIL-STD-1410, Class I and as specified herein. A fuel tank shall be furnished and shall have a capacity for not less than 10 hours operation under any operating condition specified herein. A standard 12 volt electrical system with l2-or 24-volt starting motor and heavy duty battery charging alternator shall be furnished. Batteries shall conform to SAE J537. Instrumentation and controls shall be manufacturers standard commercial items.

3.8.1.1 Air cleaner. The air cleaner shall be of the dry type conforming to MIL-A-52363, Class III. The air cleaner shall be provided with a restriction indicator conforming to MS53063-3. A removable extension tube, not less than 8 feet in length, and removable supports for mounting the extension tube in a vertical position shall be furnished. The tube shall be provided with an air-inlet hood. The opening of the tube at the airinlet end shall be covered with screen wire of not more than 1/8-inch square mesh. The cross sectional area of the tube shall be not less than 3 times the area of the air duct intake of the air cleaner. The tube and tube supports shall be stored on the crusher.

3.8.1.2 Clutch and clutch controls. The engine shall be equipped with an overcenter clutch which shall be accessible for lubrication, adjustment, and replacement of parts. The clutch shall withstand without deformation or damage repeated engagement under full load when tested as specified in 4.6.3.2. The lining shall conform to KKK-L-370, Type II and shall be installed in accordance with the lining manufacturer's instructions. Clutch control shall be accessible from the operator's platform and also accessible from the ground.

3.8.1.3 Fuel priming pump. When the outlet of the fuel tank is more than 10 feet (straight line distance) from the most distant injection pump on the engine, and electrical priming pump shall be provided and shall be mounted in the fuel tank or in the fuel line within 3 feet of the tank.

3.8.1.4 Rain cap. If the exhaust system is protected against the entry of rain by a rain cap, the cap shall be of such design that when the cap is closed and subjected to a horizontal wind from any direction, there will be a force component downward tending to keep the cap closed.

3.8.2 Electric motors. The electric motors shall be 1,800 revolutions per minute (rpm), 440-volt, 3-phase, 60-cycle, ball bearing, squirrel-cageinduction class, continuous rating, totally enclosed, fan cooled, Design C in accordance with NEMA MG1, integral horsepower, Class A insulation, temperature rise 55° C. All motors shall be provided with mounting brackets (see 3.8.2.2). Bearings shall conform to FF-B-171 and shall be a double seal type packed with grease conforming to MIL-G-23827.

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3.8.2.1 Controls, cables, and switches.

3.8.2.1.1 Power cable, reels, and ground rods. The main feeder cable shall consist of two identical 100-foot lengths conforming to MIL-C-3432, Type CO-03HLF (3/00-3/5R) 1750 or CO-03HGF (3/00-3/5R) 1750. Each length shall have a 4-pole male connector on one end and a 4-pole female connector on the other end. The cables shall have safety chains and strain collars. The safety chain shall be attached to the male end of each cable. Each connector shall have a protective cap attached with sash chain. Each cable assembly shall be equal to and interchangeable with Joy Manufacturing Company Assembly No. X8617-25. A manually operated reel or reels to contain the main feeder cable shall be permanently mounted on the crusher. The reel or reels shall be of steel construction, ratchet type, and shall include a provision to secure the cables on the reel or reels. Three ground rod assemblies for grounding the main power source shall be provided. Each ground rod assembly shall conform to W-R-550 except that the ground wire shall be 15 feet long. A 1/4-inch bronze grounding stud or bolt with nut, and two internal-external toothed washers shall be provided on the frame of the crusher near the distribution panel for attaching the groundwire clamp to the crusher. The ground rod assemblies and clamps and wires shall be stowed in the toolbox.

3.8.2.1.1.1 Power cable adaptor assembly. A power cable adaptor assembly shall be provided. The assembly shall consist of a 4-pole female connector on one end of a 5-foot length of cable conforming to MIL-C-3432, Type CO-03HLF (3/00-3/5R) 1750 or CO-03HGF (3/00-3/5R) 1750. A strain collar shall be provided to accommodate the safety chain on the male end of the main feeder cable. The adaptor shall have a protective cap attached with sash chain. The other end of the feeder cable shall have 12 inches of the cable jacket removed and shall have 1 inch of insulation stripped from each conductor. The assembly shall be equal to and interchangeable with Joy Manufacturing Company Assembly No. X86l7-24.

3.8.2.1.2 Power receptacles.

3.8.2.1.2.1 Auxiliary branch circuits. Two receptacles of the 4-pole, female type suitable for 20-ampere, 440-volt service shall be provided. The receptacles shall be equal to and interchangeable with Joy Manufacturing Co. Part No. X1697 provided with a spring-loaded cover. The inserts shall be equal to and interchangeable with Joy Manufacturing Co. Part No. 441 FIP provided with 24-inch pigtails. The receptacles shall be mounted in a bank and shall be accessible from the ground level for receiving connections from portable conveyors.

3.8.2.1.2.2 Main feeder receptacle. A main feeder receptacle shall be provided. The receptacle shall have a spring-loaded dust cover and a safety chain, and shall have a 200-ampere, 440-volt rating. The receptacle shall be equal to and interchangeable with Joy Manufacturing Co. Receptacle No. 8617-26. The main feeder receptacle shall be identified as "Main Feeder Receptacle" by an identification plate.

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3.8.2.1.2.2.1 Auxiliary main feeder outlet. An auxiliary receptacle permanently connected to and mounted adjacent to the main feeder receptacle shall be provided. The auxiliary connector shall be identical to the main feeder receptacle, except that it shall be of the female type. The receptacle shall be identified as "Auxiliary Main Feeder Outlet" by an identification plate.

3.8.2.1.3 Main feeder disconnect. The main feeder disconnect shall be a circuit breaker conforming to NEMA ICS. The circuit breaker shall be a nonautomatic circuit interrupter with shunt trip device. It shall be watertight, type IV, and dusttight, type V, in accordance with NEMA ICS. The breaker shall be 3-pole, rated for 225 amperes, and actuated by an external handle. The switch shall be labeled "OFF", "ON", and "RESET".

3.8.2.1.4 Safety cutouts. Two safety cutouts of the one-position, pushbutton, switch type, normally open and rated at 30-ampere maximum interrupting current shall be provided. The safety switch shall be watertight, Type IV, and dusttight, Type V, in accordance with NEMA ICS. The switch shall be labeled "STOP". One switch shall be accessible from the operator's platform, and a second switch shall be accessible from the ground level and adjacent to the auxiliary branch circuit receptacles.

3.8.2.1.5 Distribution panel and branch circuit protective devices. The panel shall be watertight, type IV, and dusttight, type V, in accordance with NEMA ICS. The panel shall be 3-phase, 4-wire, solid neutral, with 200-ampere main-lug capacity. Each branch circuit shall be provided with a circuit breaker in accordance with NEMA ICS. Single-pole circuit breakers will be acceptable if provided with handle tie bars for each individual polyphase circuit. Each starter shall be considered a branch circuit, and individual circuit breakers shall be provided for all starters.

3.8.2.1.6 Motor starters. Each motor on the crusher shall be provided with a motor starter. The motor starters shall be in accordance with MIL-S-12514, type I, class 2, and shall be 3-pole, alternating current, magnetic, nonreversing, across-the-line, and rated for 440-volt, 60-cycle operation. The enclosures shall be watertight, type IV, and dust tight, type V, in accordance with NEMA ICS and shall be provided with start-stop pushbuttons in the covers.

3.8.2.1.7 Internal wiring. Internal wiring shall be installed in conduit conforming to WW-C-566 or WW-C-58l, as applicable. Conduit outlets shall conform to W-C-586. A ground wire shall be carried as the fourth wire and shall be grounded to the crusher in such a manner as to assure an electrical bond between all electrical apparatus and the crusher on which it is mounted. Branch circuit conductors, including wiring internal to starters, breakers, and other electrical wiring and devices which carry motor-running current, shall have a current-carrying capacity of not less than 125 percent of the applicable motor-running current. Control subcircuits not carrying motor-running current shall be wire not smaller than No. 14. Such subcircuits shall include wiring internal to the control panel but not internal to starters, breakers,

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and other electrical devices. Plug-in-type connections in branch circuits shall have a positive lock or other effective means to prevent accidental disconnection caused by vibration. The contractor shall furnish a wiring diagram showing each wire in the system number coded near each connector.

3.8.2.2 Motor-mounting brackets. All motors shall be provided with mounting brackets. When a 5-horsepower (hp) motor is specified herein, the mounting brackets shall also be of sufficient size and shall be provided with mounting holes to accommodate a 10-hp motor for emergency replacement.

3.9 Drives.

3.9.1 Chain drives. Unless otherwise specified herein, chains used in drive trains shall be in accordance with ANSI B29.1 and shall conform to MIL-STD-423 and MIL-STD-424 as applicable. Chain drives shall be installed with idlers and other methods for providing adjustment.

3.9.2 V-belt drives. V-belt drives shall be in accordance with ANSI B55.l or Rubber Manufacturers Association, Inc., "Engineering Standard for Drives Using Narrow V-Belts (3V, 5V; and 8V Cross Sections)". Belts used on multiple V-belt drives shall be matched sets or banded together by a tie band. Jack shaft drives may be used. All V-belt drives shall have provisions for adjustment.

3.9.3 Gear drives. Gear drives shall be totally enclosed and shall run in an oil bath. Cast iron gears when used shall conform to ASTM A536. The analysis and heat treatment shall be as recommended by the contractor. When cast iron gears are provided, the pinion shall be of steel.

3.9.4 Shafting. Shafting 2 inches in diameter and larger shall be of steel in accordance with ASTM A193, type B7, Rockwell hardness 25 to 32 on Rockwell "C" scale and shall be turned, ground, and polished. The fit tolerance between all shafting and driven members shall not exceed 0.003 inch.

3.9.5 Keys and keyways. All keys and keyways on the crushing component (see 3.12), except where tapered hubs or taper-lock bushings are used, shall be in accordance with Figure 1. Keys shall be at least as hard as the shafting.

3.9.6 Torque arm speed reducers. On torque arm speed reducers, the torque arm shall be attached to the speed reducer housing by means of one or two brackets. Each bracket shall be attached to the speed reducer housing by not less than two bolts. The torque arm shall be at an angle of 90 degrees plus or minus 30 degrees with a line drawn between the output shaft and the point at which the torque arm is attached to the speed reducer. If the torque arm is used to tighten the drive belt or chain, the drive shall be at an angle of 90 degrees plus or minus 10 degrees with a line drawn between the input and output shafts of the speed reducer. The angular measurements specified herein shall apply when the drive belt or chain is properly adjusted for tension. Concentric loading shall be provided on speed reducer housing when the torque arm is used for drive belt or chain tension adjustment.

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3.10 Feeder. The vibrating grizzly shall be fed by a feeder not less than 3 feet wide by not less than 8 feet long. The feeder shall receive quarry run rock to include individual rocks weighing not less than 1000 pounds dumped from power shovels without, breakage of parts, damage to feeder or components, or reduced production. If an apron type feeder is furnished, the feeder shall have overlapping steel pans or flights mounted on not less than three hardened alloy steel roller chains supported on a series of load-carrying rollers or rails. The chain shall conform to ANSI B29.1. The maximum design load shall not exceed 15 percent of the minimum ultimate tensile strength of the chain. All bearings on the feeder shall be fitted with grease fittings. The feeder shall be equipped with a 10-hp electric motor conforming to 3.8.2. A receiving hopper not less than 8 feet wide and not less than 9 feet long shall be mounted on the feeder. The feeder hopper shall be provided with side plate extensions over the vibrating grizzly so as to deliver material from the feeder to the vibrating grizzly without spillage. The hopper shall be fabricated from steel plate conforming to ASTM A242 having a thickness of not less than 3/4-inch with no liners required. To provide roller chain tension, a take-up adjustment consisting of two Acme thread arrangements (one per side), with threads conforming to ASNI B1.5, shall be provided at the tail shaft.

3.11 Grizzly. The crusher shall be equipped with live type grizzly not less than 3 feet wide and not less than 5 feet long, having two level or sloping decks and equipped with bypass chutes. The grizzly shall be provided with a 10-hp electric motor conforming to 3.8.2. The top deck shall consist of heavy duty, high-carbon-steel bars or rails extending the full length of the grizzly. The bars or rails shall withstand the discharge of heavy abrasive rock from the feeder and shall be spaced to permit passing 2-inch-minus rock onto the lower deck. The second deck shall be equipped with a 3 by 5 foot, 3/4-inch slotted screen cloth to screen off one size and discharge the oversize onto the end discharge conveyor belt. A hopper shall be provided below the second deck. The hopper shall contain a flop gate or means of bypassing material either from the hopper to the end discharge conveyor; or into a discharge chute or side delivery conveyor as specified in 3.13.12. When a discharge chute is furnished, it shall be provided with means for diverting materials to either side of the crusher and shall provide ground clearance for the receiving hopper of a portable conveyor.

3.11.1 Screen activating mechanism. Any screen activating mechanism subject to material from the deck or decks striking on the mechanism shall be equipped with a replaceable, protective cover.

3.11.1.1 Shaft bearings. All shaft bearings shall be double-row, selfalining, antifriction type. All bearings exposed to dust shall be provided with labyrinth-type grease and dust seals which will prevent dust from entering the bearings. The drive shaft and bearings shall withstand the loads imposed under any operating condition specified herein.

3.11.2 Woven wire screen cloth. Screen cloth shall conform to MS39127.

3.11.2.1 Material. All screen cloth shall be made of crimping grade, oil-tempered wire conforming to NSGA Circular No.80, table 3, grade B, carbon range 0.55 to 0.70, drawn and hardened as specified.

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3.11.2.2 Wire size. Wire shall conform to NSGA Circular No. 80, table 7, standard heavy, and shall be sized as follows:

Clear opening, minimum Inch	Wire diameter Inch
3/4 by 4	0.250
1/2 by 4	0.192

Two 3 by 5 foot screen cloths of each wire size shall be furnished with each crusher in addition to the 3/4-inch cloth installed on the second deck of the grizzly.

3.11.2.3 Weave. All screen cloth shall have slotted openings with slots parallel to the hook strips, and shall be of double-crimp, construction. Tension shall be applied from the sides, there shall be no crimping in the longitudinal wires of the 4-inch openings.

3.12 Crushing component. The crushing component shall be of the jaw type with overhead eccentric action. The opening of the jaws measured at the top shall be not less than 20 by 36 inches at nominal 3-inch closed setting. The jaws shall be adjustable for a closed setting from a minimum of not more than 2 inches to not less than 5 inches with the toggle plate furnished. The jaw-crushing component shall be powered by an integrally mounted diesel engine which shall conform to 3.8.1. The crushing component shall crush materials ranging in size from a cross section of 18 by 34 inches down to the size of the minimum jaw setting.

3.12.1 Frame. The crushing component frame shall be constructed of cast steel or welded steel plate. Welded steel plate shall be stress relieved after welding in accordance with American Welding Society codes for weldments of this size. Cast steel, when used, shall be in accordance with ASTM A148, with analysis and heat treatment as recommended by the manufacturer of the crushing component. The contact points on the frame which support the parts that transmit the crushing stresses to the frame shall be machined, or provisions shall be made to distribute the crushing load on the frame so that the frame will safely withstand the stresses imposed when crushing maximum size material.

3.12.2 Jaw and cheek plates. Jaw and cheek plates shall be made of cast manganese steel in accordance with ASTM A128, with analysis and heat treatment as recommended by the manufacturer of the crushing component. The movable and stationary jaw plates shall be reversible and shall be secured so that they will not loosen and shift during the crushing operation. The back surfaces of the jaw plates shall be in a plane so that they will fit against their respective supporting member and provide equal load pressures in operation. The jaw plates shall be not less than 3 inches thick at the main crushing area measured from the back bearing surface of the plate to the valley of the corrugations, and shall be not less than 2 inches thick at the top and bottom of the plate measured in the same manner. Corrugations shall be not

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less than 3/4 inch deep measured from crest to valley. The jaw plates shall be removable for replacement without removal of other accessories or components except cheek plates and wedges. The jaw plates shall have inserts for attaching lifting eyes or similar means for removing or inserting new jaw plates. Cheek plates shall be mounted on the frame between both sides of the jaw plates to protect the frame during the crushing operation. The cheek plates shall be constructed of two or more replaceable pieces and shall not loosen and shift during the crushing operation.

3.12.3 Pitman. The pitman shall be made of cast or fabricated steel reinforced with ribs and shall be machined to receive the movable jaw plate. Cast steel when used shall be in accordance with ASTM A148 with analysis and heat treatment as recommended by the manufacturer of the crushing component. Fabricated steel, when used, shall be stress relieved in accordance with American Welding Society codes for weldments of this size. At the lower, or toggle end of the pitman, provision shall be made to receive a renewable toggle seat.

3.12.4 Pitman shaft bearings. The pitman and pitman shaft shall be mounted on antifriction bearings. The bearings shall be protected by labyrinth-type grease and dust seals which shall prevent dust and other foreign matter from entering the bearings. The bearings shall conform to FF-B-185 or FF-B-187, except that taper bore bearings and taper sleeve adaptors will be permitted.

3.12.5 Pitman shaft. The pitman shaft shall be an eccentric shaft machined from steel conforming to SAE 4140, 4340, or 6150, heat-treated, and shall withstand the maximum crushing load. The main shaft bearings shall be mounted in bearing housings on the crusher frame.

3.12.6 Flywheels. Two flywheels of gray iron conforming to ASTM A48, Class 35, 40, 45, 50, 55, or 60 shall be mounted on the pitmin shaft, shall be counterbalanced to offset the mass of the eccentric revolving parts, and shall be of sufficient size to provide inertia to transmit and equalize the energy necessary for crushing the rock specified in 3.1 at maximum crusher capacity, without reducing the flywheel speed by more than 10 percent from the flywheel speed when the crusher is operating empty. One flywheel shall be machine-grooved for V-belts, and used as a driving pulley.

3.12.7 Toggle plate. A cast iron or steel toggle plate shall be provided to rest between the toggle seat on the pitman and the seat in the toggle wedge of the adjusting mechanism. The toggle plate shall be held in place by means of a tension bolt and shall be spring arranged so that the bolt will not bind or bend during operation and adjustment. The toggle plate shall be constucted so that it will break or deform when uncrushable material such as tramp iron enters the crushing component, and shall fail before any other part of the crusher is stressed so as to cause breakage or permanent deformation. The toggle plate shall not break when crushing rock as specified herein.

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3.12.8 Adjusting mechanism. An adjusting mechanism of the wedge-and-screw or shim type shall be provided which will permit adjustment of the opening between the jaws. The adjusting mechanism shall be cast steel or fabricated steel plate, and shall be reinforced to withstand crushing of maximum size material. All contact surfaces of the adjusting mechanism shall be machined to insure uniform distribution of the load on the respective parts of the crushing component frame. A stop shall be provided to prevent adjustment to actual contact of the jaws during any part of the crushing cycle.

3.12.9 Jaw setting. The jaw setting shall be measured along or parallel to the longitudinal axis of the jaw crusher on a horizontal plane. Measurement shall be of the greatest opening in the minimum discharge position of the jaws in the crushing cycle, normally between the ridge of one jaw and the valley of the other. Variation between any two measurements for any jaw setting shall be not greater than 1/4-inch. Jaw settings shall be provided which can produce the materials specified in 3.12.

3.13 Conveyors. The crusher shall be equipped with an end-delivery conveyor which shall receive materials from the grizzly chutes and jaw crushing component in a receiving hopper. The conveyor shall have a belt 30 inches wide with a belt speed not less than 300 feet per minute. The conveyor shall have not less than 10 feet of ground clearance measured on the vertical from the centerline of the head pulley.

3.13.1 Frame. The conveyor frame shall be of rigid construction, shall support all other components of the conveyor not supported directly by the chassis, and shall withstand all static and dynamic loads imposed by other components when operating with a loading of 50 pounds per lineal foot of belt or when traveling. When hinged sections of frame are provided, stiffeners, braces, or other means shall be provided to maintain frame rigidity during operation and travel.

3.13.2 Head pulley. The head pulley shall be the drive pulley, shall be of steel construction not less than 32 inches wide, not less than 14 inches in diameter, and shall be crowned and lagged. The crown shall taper 1/8-inchper-foot to the center of the pulley. The outside surface of the pulley shall be concentric with the shaft within a total indicator reading of 0.060 inch, except a total indicator reading of 0.090 inch is permissible in an area immediately adjacent to a longitudinal weld.

3.13.3 Tail pulley. The tail pulley shall be of steel construction not less than 32 inches wide, and not less than 12 inches in diameter, shall be the wing or self-cleaning type, and shall be crowned. The crown shall taper 1/8-inch-per-foot to the center of the pulley. The outside surface of the pulley shall be concentric with the shaft within a total indicator reading of 0.100 inch.

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3.13.4 Taper-type locking device (pulley and shaft). The head and tail pulleys shall be provided with an inner hub which clamps to the shaft by moving axially along a taper in the outer hub. The mechanical clamping shall be accomplished by the use of a screw member or members providing for mounting and removal of the pulley.

3.13.5 Bearings. The head and tail shafts shall be mounted in self-alining, antifriction, greasable-type bearings furnished with labyrinthtype grease and dust seals. Bearings shall conform to FF-B-185. Mounting of bearings in a manner other than securing against shoulders shall be subject to the approval of the contracting officer for the specific application.

3.13.6 Takeup adjustment. To provide belt tension, a takeup adjustment consisting of two l-inch Acme thread arrangements (one per side) shall be provided at the tail pulley. The total available belt take up shall be not less than 12 inches.

3.13.7 Belt carriers. The conveyor belt shall be carried between pulleys on troughing rolls, troughing impact rolls, and return rolls conforming to the Conveyor Equipment Manufacturers Association (CEMA) Standard No. 502.

3.13.7.1 Troughing rolls. The troughing roll assemblies shall be spaced not more than 3 feet apart and shall start from the last impact roll and continue toward the head end of the upper run.

3.13.7.2 Impact rolls. The area under the receiving hopper, including the area under the jaw crushing component, shall be provided with impact type troughing carrier rolls mounted at intervals of not more than 1 foot measured center to center.

3.13.7.3 Return rolls. The return rolls and mountings shall be spaced not more than 6 feet apart. One return roll shall be mounted on the upper run of the conveyor ahead of the tail pulley and underneath or slightly to the rear of the back flashing on the receiving hopper. The mounting may be modified to suit the installation. If the vertical centerline of the tail pulley is within 6 inches of the back flashing on the hopper when 6 inches of belt takeup remains, the return roll shall not be required.

3.13.8 Discharge conveyor belt. The discharge conveyor belt shall be not less than 30 inches in width, and constructed with not less than two plies of synthetic tensile fabric. The top rubber cover shall be 1/8 inch thick and the bottom cover shall be 1/16 inch thick with a skim coat between all plies.

3.13.8.1 Belt fasteners. The conveyor belt shall be provided with bolted-metal-plate fasteners. If the conveyor belt is removed for shipment, it shall be provided with bolted-metal-plate, hinge-type fasteners capable of disassembly and reassembly by removing and replacing the hinge pin.

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3.13.9 Driving mechanism. The driving mechanism shall consist of a reducing gear arrangement powered by a 10-hp electric motor (see 3.8.2) through a multiple V-belt drive. The reducing gear shall be totally enclosed and operated in an oil bath. Bearings shall be antifriction type and shall conform to FF-B-171, FF-B-185, or FF-B-187, as applicable. Bearings conforming to FF-B-171 may be retained on the shaft by a bearing-retainer plate and shims between the retainer plate and housing to provide endwise adjustment.

3.13.9.1 Holdback device. An automatic holdback device of the overrunning, clutch type shall be furnished to prevent slippage or reversal of flow when the motor stops.

3.13.10 Belt cleaners. A weight-loaded or spring-loaded, replaceable, single-rubber-blade belt scraper shall be mounted at the drive (head) end pulley. The scraper shall cause the scraped material to fall forward and toward the main discharge. The blade shall be neoprene rubber not less than 1 inch thick and shall be split in the center to match the contour of the crown. A weight-loaded or spring-loaded, V-shaped, rubber-blade scraper shall be mounted inside the return run of the belt at the foot (tail) end pulley. The blades shall be neoprene rubber and shall be not less than 3/8-inch thick. The scrapers shall exert uniform pressure across the full width of the belt.

3.13.11 Receiving hopper. The receiving hopper shall be mounted at the lower end of the conveyor frame, shall be removable, shall be of the skirt board type, and shall be three-sided. The hopper shall have sloping sides to prevent spillage of material. The hopper shall extend not less than 12 inches beyond the discharge of the crusher jaws toward the drive end of the conveyor. The space between the three sides of the hopper and the conveyor belt shall be provided with adjustable and replaceable rubber flashing mounted under the hopper plates. The hopper plates shall be of steel conforming to ASTM A242 and shall be not less than 0.187 inch in nominal thickness.

3.13.12 Side delivery conveyor. When a side delivery conveyor is furnished (see 3.11), it shall be driven by a 5-hp motor as specified in 3.8.2. Means shall be provided for delivery of material to either side of the jaw crusher. A reversing switch may be used to reverse the direction of the discharge, and a warning plate shall be provided to guard against reversing while the motor is running. The discharge ground clearance shall be of sufficient height to permit delivery of material into the receiving hopper of a portable conveyor.

3.14 Trailer components. The physical characteristics of the crusher trailer in operational condition for processing rock, with dolly, all components, attachments, accessories, equipment, tools; and fuel, oil, and coolant reservoirs filled to rated capacity shall be as specified below. All measurements shall be made with the crusher on level ground, and the inflation pressure of the tires shall be as recommended by the tire manufacturer for operation at 30 mph highway speeds. Axle loads shall be measured at the ground under the tires. Published component and vehicular

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ratings and capacities shall not be increased to meet the requirements of this specification. The dolly drawbar shall be in the straight-ahead towing position. The minimum practicable number of components may be removed to meet the reducible height specified in (f) below.

(a)	Total weight	80,000 lb., max.
(b)	Fifth wheel load limit	30,000 lb., max.
(c)	Axle load limit, each	18,000 lb., max.
(d)	Overall length	600 in., max.
(e)	Overall width	120 in., max.
(f)	Reducible highway transport height	156 in., max.
(g)	Ground clearance (SAE J894)	9.5 in., min.
(h)	Ground clearance (excluding axles) with dolly and with either truck-tractor M9l6 or M920	12 in., min.
(i)	Upper fifth-wheel height	64 in ⇦ 2 inches
(j)	Stabilizing jack (front) clearance radius ("LWC" of SAE J70l)	88 in., min.
(k)	Swing radius ("SR" of SAE J70l)	58 in., max.
(l)	Kingpin setting range ("KP" of SAE J70l)	21 through 36 in.

3.14.1 Frame. The trailer frame shall be constructed to withstand, without damage or permanent deformation, the static and dynamic loads imposed by operation and by being transported as specified in 3.17.

3.14.1.1 Upper fifth wheel and kingpin. The front of the semitrailer shall be constructed so as to be compatible for coupling with and towing by both the military M920, 20-ton truck-tractor and the M9l6, trucktractor (see 3.31). The crusher shall be furnished with a removable, auxiliary fifth wheel adaptor as specified herein. The upper fifth wheel plate shall be of the full-width-type that permits coupling at any angle up to 75 degrees from straight ahead, and shall cover a 36-inch-diameter lower fifth wheel (see 3.15.1). The upper fifth wheel assembly shall conform to DOT regulation 293.70. The plate shall be high tensile steel (not less than 50,000 psi yield point) and shall be not less than 3/8 inch thick. The plate shall include front and side coupling ramps, having not less than 1-1/4 inch vertical height. Unless the coupling ramps are protected by a front crossmember and the right and the left frame rails or structural members, drain holes shall be provided to prevent any collection of water. When selected, the auxiliary upper fifth wheel adaptor coupled to the crusher shall meet all of the requirements for the upper fifth wheel of the crusher, including coupling ramps. The upper fifth wheel of the crusher shall be equipped with an SAE J848 (3-l/2-inch) kingpin for the M9l6 and M920 truck-tractor compatibility and the adaptor shall be equipped with an SAE J700 (2-inch) kingpin for the other truck-tractor compatibility. The level crusher fifth wheel heights, with and without the adaptor, shall coincide with the applicable truck-tractor height as specified in 3.14. The coupled adaptor shall remain

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stationary with respect to the crusher during all transport operations. The adaptor shall be equipped with positive manually operated locks and fasteners and with operational instructions that shall permit two men to install and remove the adaptor with common tools in not more than 1 hour. Retaining brackets shall be provided on the crusher for stowing the auxiliary fifth wheel adaptor when not in use.

3.14.1.2 Rear towing components. Towing components shall be furnished on the rear of the semitrailer as specified and in accordance with SAE J849 for doubles trailer application, except as specified herein.

3.14.1.3 Front and rear intervehicular connections. The semitrailer shall be furnished with airbrake couplings specified in 3.14.5 and electrical receptacles specified in 3.14.6 for connecting (1) the truck-tractor to the semitrailer, (2) the dolly to the semitrailer and (3) other trailers to the semitrailer. The front connections shall comply with SAE J702 and the rear connections with SAE J849. Service brake couplings shall be on the curb side, color coded blue, and emergency brake couplings shall be on the road side, color coded red. Clearance about the couplings and receptacles shall allow operating personnel to connect the truck-tractor, or dolly hoses and harness to the semitrailer, and from the semitrailer to towed trailers without interference with other components. Each airbrake coupling shall be protected with a spring-type, metal, dust cover. The front electrical receptacle shall have leads terminating within a waterproof, electrical, semitrailer-type commercial nosebox or junction boxes.

3.14.2 Rear suspension. The rear of the semitrailer shall be mounted on single wide base wheel and tire, multiple-axle suspension system. The suspension manufacturer's rated capacity of the suspension assembly shall be not less than the loading imposed on the suspension, measured at the ground, with all components of the crusher assembled for rock crushing operation. The suspension shall be of the equalizing type, shall be equipped with elastomer bushings or blocks that eliminate the need for lubrication, shall have elastomer bumpers at the limit of articulation, and shall provide freedom from brake hop. Each axle shall be rated by the axle manufacturer at a commercial highway usage, capacity rating of not less than 25,000 pounds and shall be loaded for this application not to exceed that specified in 3.14 when measured at the ground. Bearing cones, bearings, and bearing cups shall be furnished by the same bearing manufacturer. The suspension shall be in correct alinement with the frame and with the axles so that the diagonal measurements from the true center of the kingpin to the outer center of each spindle on the axle do not differ in excess of 1/8 inch for every 20 feet or fraction thereof in diagonal distance. Each subsequent axle shall be parallel to the front axle of multiple axle suspensions such that the axle wheelbases on each side do not differ in excess of 1/8 inch. Axle alinement measurements specified herein shall be made in accordance with SAE J875 or by other methods of equal accuracy. Multiple axle wheelbases shall be not less than 50 inches center-to-center. Suspension mounts shall be installed on the frame and the axles in accordance with the suspension and axle manufacturer's recommendations.

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3.14.3 Wheel assemblies. The semitrailer shall be equipped with single wide base disc wheels size 22.5 x 14:00 conforming to SAE J876. The wheel shall be equipped with 18 x 22.5, 16-ply rating, wide base tubeless tires having traction-lug tread design. The semitrailer shall be furnished with one spare wheel with tire, securely mounted, and shall not interfere with any component of the crusher during travel or operation. The spare shall be located to permit removal and replacement by not more than two men. The brake drums shall have a wearing surface of SAE J43l automotive gray cast iron, SAE G3500 or SAE G4000, and shall be of the antibellmouth type. Wheel assemblies including hubs and brake drums shall be inherently balanced.

3.14.4 Chock blocks. Two chock blocks with captive chains and two wire cradles shall be furnished. The captive chain shall be wound in a figureeight manner around the cradle ears and the hook snapped to the lower cradle loop. The length of the captive chain shall permit the clock blocks to be placed ahead of or behind the semitrailer tires. The chock blocks and cradles shall conform to Figures 2 and 3 respective. Installation of the cradles shall be with the two cradle ears uppermost and the short loops fastened to an approximately vertical, flat surface.

3.14.5 Brake system. The plant shall be equipped with an immersion resistant air-brake system in accordance with DOT Federal Motor Carrier Safety Regulations (FMCSR) and DOT Federal Motor Vehicle Safety Standards (FMVSS), as applicable except as otherwise specified herein. The brake system shall include service and emergency couplings (front and rear), relay emergency valve, air reservoir, cylinders or chambers, foundation brake assemblies on each wheel, and all other components and plumbing required for a complete, matched air brake system. All brake system components and plumbing shall be located, installed, and protected to prevent damage or malfunction from ground obstructions and debris when transported, from falling aggregate when operating and to obtain maximum efficiency and ease of maintenance. Service and emergency couplings shall be the noninterchangeable types, color coded, blue for service and red for emergency, with a spring-type metal dust cover installed on each coupling to prevent debris from entering the openings. Metal identification plates conforming to MS53007 shall be installed near each front and rear fixed coupling. The air reservoir shall conform to SAE J10 and shall have an air storage capacity of not less than two times the recommended air volume specified for truck trailers in SAE J813. The reservoir shall be equipped with a remote or manual controlled 90 degree twist drain valve mounted on the low end of the reservior. The brake assemblies shall be equipped with brake lining selected by the contractor and shall be a standard size as listed in the Automotive Data Book of the Friction Materials Standards Institute, Inc. The lining shall be installed on the shoes by use of hardened steel bolts and nuts as recommended by the lining manufacturer. All metal surfaces enclosed by the brake drum and the dust shields shall be coated with a black oxide primer. Wedges, pins, cams, rollers, and anchors shall be coated with a preservative oil.

3.14.5.1 Brake lines. The plumbing of the brake system shall consist of flexible air lines having reusable fittings. The brake lines shall be routed within the protection of the semitrailer framing and shall be secured by cushioned support clamps spaced at intervals not greater than 3 feet. Through-frame connections or grommets used in conjunction with clamps shall be

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required where lines pass through sheet metal or framing members. The routing and connection locations shall facilitate replacement of the lines from bulk supply. Not less than 5 inches of slack in flexible air hose shall be provided at each of the relay emergency valve connections. The hose ends of these lines shall have identification markers near each fitting. Brakeline components shall conform to the following:

(a) Intervehicular air hose (on dolly converter); MIL-H-3992, type I, class l or 2, with couplings and dust covers on each end.
(b) Flexible air hose; SAE J40, SAE 40R2, type E or type F, having reusable fittings.
(c) Intervehicular couplings; SAE J3l8, polarized noninterchangeable type with dust covers.
(d) Identification tags; MS53007.

3.14.6 Vehicular electrical system. The crusher shall be equipped with a waterproof, 12-volt vehicular electrical system for service and blackout night travel. All components and wiring shall be so located and guarded to provide protection from vibration, breakage, and damage from falling aggregate. Except as specified herein, the electrical system shall be in accordance with the following requirements as applicable for this type equipment:

(a) SAE Handbook, Electrical Equipement and Lighting Section.
(b) DOT Motor Carrier Safety Regulations.
(c) TTMA Recommended Practice RP#9 and RP#14
(d) Figure 4.

3.14.6.1 Military vehicle electrical system. When specified (see 6.2), the trailer shall be equipped with a military compatible 24-volt service lighting system to provide towing capability for the military design trucks as well as commercial trucks. In addition to the TTMA 7-contact plug, the trailer shall be equipped with the following military intervehicle components:

(a) Military 12-Contact Plug – MS75020-1
(b) Military Intervehicular Cable – MIL-C-13486/12.

The military plug and cable shall be the same length as the TTMA plug and cable and both shall be labeled with identification plates for 24-volts and 12 volts respectively, negative ground. The military 24-volt intervehicular cable shall be routed into a weatherproof stud and nut type junction box, equipped with SAE J553 circuit breakers, protecting live circuits B, E, and J of the military service lamp portion. All unused leads of the cable shall be insulated and sealed watertight. The military 24-volt circuits shall be connected through replaceable resistors, for live circuits, and to the trailer frame, for ground circuits, in accordance with table III.

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TABLE III. Military-commercial connections.

Function	Mil. Plug Terminal	Connection Instructions	Comm. TTMA Wiring Ident.
Blackout Marker	A	No.	-
Ser. Left Turn and Stop	B	Connect to –	YEL (3)
Blackout Marker	C	No.	-
Ground	D	Connect to –	WHT (1)
Ser. Tail & Marker	E	Connect to –	BLK (2)
		& Connect to –	BRN (6)
Blackout Stop	F	No.	-
Blackout Marker	H	No.	-
Ser. Right Turn and Stop	J	Connect to –	GRN (5)
Aux. Power(Hot)	K	No.	BLU (7)
Ground	L	Ground to Frame	-
Electric Brake	M	No.	-
Electric Brake	N	No.	-

With the military system installed, the TTMA marker circuits, color coded BLK (2) and BRN (6), shall be electrically isolated from each other, thus preventing 12-volt current cross-over when operated by commercial trucks through the 12-volt TTMA plug. Replaceable diodes may be used, having a minimum amperage of 150 percent of the current demand of the running lamps served. Military circuits letter coded B and J on the tactical trucks are passenger car combination stop and turn signal type, using the same lamp filament, and there is no single stop lamp circuit. As a result, the trailer's turn signal lamp circuits, YEL (3) and GRN (5), shall function as both turn signal lamps and stop lamps and the trailer's normal stop circuit, RED (4), cannot be energized from the military 24-volt system input.

3.14.7 Stabilizing jacks. The semitrailer shall be equipped with not less than four screw-action stabilizing jacks permanently attached to the chassis. The jacks shall conform to GGG-J-51, Type I, Class 3, rated capacity 15 tons, except that the diameter of the screw shall be not less than 2-1/2 inches. The jacks shall have steel ball-bearing caps. Detachable pads shall be furnished. Each jack pad shall have sufficient area to reduce ground pressure to less than 100 psi. Provisions shall be made for stowing the pads in the toolbox. The stabilizing jacks shall fold and be secured for travel. The two front stabilizing jacks shall also serve as landing gear and support the front of the crusher when there is nothing under the upper fifth wheel. The stabilizing jacks shall be located so as to stabilize the crusher when the crusher frame is level, dolly is installed under upper fifth wheel, and all wheels are blocked up clear of the ground. Two 20-ton hydraulic jacks shall be furnished to raise and lower the crusher as required for coupling and uncoupling the dolly or truck-tractor. The hydraulic jacks shall conform to GGG-J-51, Type VII, Class 2, Style B. The hydraulic jacks shall be stored in the toolbox. When specified, the jacks will be furnished to the contractor by the Government.

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3.15 Dolly. The dolly shall be furnished by the contractor.

3.15.1 Contractor-furnished dolly.

3.15.1.1 Fifth wheel. A fixed lower fifth wheel shall be furnished. The height of the fifth wheel with the semitrailer loaded shall be 61 inches plus or minus 2 inches.

3.15.2 Drawbar coupler. A ring drawbar coupler in accordance with MS51337 shall be installed in a housing integral with the towbar. The housing shall support the coupler the entire length of the shank. The clearance between the shank and support shall be not greater than 0.035 inch. When the dolly is in towing position, the height range of the lunette shall provide compatibility with pintles in SAE J849, ranging between 31 through 37 inches from level ground. The center of the lunette and the center line of the towbar shall be located within 1/8 inch of the longitudinal center line of the dolly. The towbar shall be hinged. When the dolly is loaded and the towbar is not locked rigid, the hinge shall permit the lunette to lower without additional load. The towbar shall be rigid when the dolly is being stored or towed without load. When coupled to the semitrailer, the dolly shall turn 110 degrees both to the right and to the left of the longitudinal centerline of the semitrailer. The coupler shall be compatible for towing with a military, M917 20-ton dump truck or with a military 20-ton, 6 x 6, M9l6 truck-tractor and shall permit the centerline of the truck to be at an angle of not less than 60 degrees to the right or to the left of the centerline of the dolly when in towing position. There shall be no interference between either the truck or the dolly-semitrailer combination.

3.15.3 Landing leg. The towbar of the dolly shall be provided with an adjustable landing leg to permit one man to couple and uncouple the lunette and the pintle on the towing vehicle. Ground clearance shall be not less than 18 inches when the towbar and landing leg are in travel position.

3.15.4 Dolly suspension. The dolly suspension shall include two single wide base tire axles as specified in 3.14.2 and 3.14.3 in a single point mount, tandem axle configuration. A complete air brake system shall be furnished as specified in 3.14.5. The dolly brakes shall be correctly balanced in application pressure and timing to reduce the possibility of jackknifing the crusher during stops. The dolly brakes shall have less effectiveness than the rear crusher brakes through use of smaller chambers or by installation of a pressure holding or reducing valve in the dolly brake system.

3.15.5 Dolly electrical system. The electrical system of the dolly shall conform to 3.14.6. Service and blackout rear combination lights only shall be required and installed.

3.15.6 Dolly intervehicular connections. The dolly shall be provided with electrical and airbrake lines to connect the truck systems to the semitrailer systems. All forward intervehicular connections between the dolly and the truck shall extend 36 to 42 inches toward the truck and

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terminate on the dolly towbar rearward at least 20 inches, measured from the center of the lunette eye. All connections during temporary storage and travel when not in use. Electrical and airbrake components shall be as specified in 3.14.1.3 and shall include hose spring guards and dust covers on the airbrake rearward intervehicular connections between the dolly and the semitrailer shall be supported on a truck-tractor-type spring pole. The intervehicular connection shall have a minimum length of 10 feet, measured from the dolly frame to the semitrailer connectors and shall allow the dolly to turn as specified in 3.15.2. Plug clips and rigid mount dummy couplings shall be mounted on the dolly and spring pole to hold both the forward and the rearward intervehicular coupling hose ends. Forward, intervehicular connections shall be routed so that dropping of the tailgates of the military dump and cargo trucks shall not cause damage to the connections.

3.15.7 Safety chains. Two safety chains fitted with grab hooks shall be furnished. The safety chains shall extend 36 to 42 inches beyond the center of the drawbar lunette eye. The safety chains shall be in accordance with DOT Motor Carrier Safety Regulations, except that the safety-chain hooks shall fit the shackles and attachments of the towing vehicles and of the semitrailer (see 3.14.1.2).

3.16 Operator's platform. The crusher shall be provided with a steel platform which shall extend across the front and both sides of the jaw crusher opening. The platform shall be not less than 7 feet in width and not less than 5 feet in length and shall be equipped with hand rails. All platform flooring shall be made of expanded metal. The platform shall permit access to the crusher opening and feeder.

3.16.1 Auxiliary components. Access ladders, platforms, and handholds shall be provided to permit servicing and inspection of all components. An auxiliary steel platform shall be furnished for servicing the engine.

3.16.2 Controls. The engine clutch control shall be located on the operator's platform and shall be of the mechanical positive type. Additional provisions shall be made to provide clutch control from the ground level. Controls for the feeder, grizzly, and conveyors shall be located on the operator's platform.

3.17 Transportability. The crusher shall withstand impact forces encountered in vehicular, rail, and air transport as specified herein without damage or permanent deformation.

3.17.1 Vehicle mobility. The crusher shall meet the performance characteristics when coupled to its dolly converter and when towed by a prime mover, such as the 20-ton M917 dump truck or crawler tractor(s), and when directly coupled as a semitrailer to a 20-ton, 6 x 6, M9l6 truck-tractor and to a 20-ton, 8 x 6, M920 trucktractor.

3.17.1.1 Highway operation. The crusher shall travel over smooth, hardsurfaced primary highways at speeds up to 30 mph.

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3.17.1.2 Cross-country operation. The crusher shall negotiate unimproved roads and open, rolling, and hilly cross-country terrain at speeds up to 10 mph. The crusher shall traverse grades up to 40 percent; mud, sand, and snow; and transport incident to intended operation sites, when towed by crawler tractors. Without prior preparation, the crusher shall ford hardbottomed bodies of water up to 30 inches in depth without damage or adverse effects because of immersion.

3.17.1.3 Trailing and turning ability. The crusher shall trail the towing vehicle without weaving or sidesway. The crusher shall follow the truck-tractor without exceeding 75 percent of the DOT tracking deviation, as specified in 293.70 of the DOT Regulations, when towed over a paved surface. Jackknife turns up to an angle of 90 degrees for truck-tractor semitrailer combination and turns up to 60 degrees for truck-dolly semitrailer combinations as measured from straight-ahead to the right and to the left shall not cause damage to or interference between the prime mover, dolly, or crusher.

3.17.1.4 Braking ability. The combined service brakes of the prime mover and crusher shall control, decelerate, and stop the combination in accordance with DOT Regulations, except that the brake application and stopping distance shall be accurately measured from initial movement of the brake pedal to the stopping point. The emergency breakaway system shall provide a controlled, emergency brake application that shall stop and hold the crusher without hazardous unbalance when actuated by disconnect, breakage, or leakage of the emergency line. With the brake system of the crusher stabilized at maximum working pressure of the coupled prime mover, the air leakage within the semitrailer brake system shall not exceed 1.0 psi per minute with the brakes released, nor exceed 2.0 psi per minute with the brakes applied.

3.17.1.5 Suspension system. The suspension systems of the dolly and crusher shall provide a nondamaging stable ride for highway travel and the articulation specified herein for cross-country travel. The suspension system shall withstand the imposed loads without evidence of overload, wear, malfunction, or permanent deformation when they are subjected to the loading and the operating conditions specified herein. When the dolly is coupled to the semitrailer, the articulation shall provide not less than 8 inches difference in elevation between opposite wheels on each axle without loss of tire-to-ground contact. In addition, not less than 8 inches articulation shall be required between all combinations of diagonally opposite wheels on multiple axle suspensions without loss of tire-to-ground contact. Eight inches of articulation shall be 4 inches up and 4 inches down.

3.17.2 Railroad. Sectionalization for rail transportation shall be within the dimensional limits shown on Figure 5, except that an item length of 38 feet 10 inches in permissible. Sectionalization and reassembly shall each be accomplished in not more than 6 hours without cutting or welding.

3.17.3 Air transport. Sectionalization for air transport shall be within the dimensional limits shown on Figure 6. Sectionalization and reassembly shall each be accomplished as specified in 3.17.2.

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3.18 Lifting and tiedown attachments. All sections separated to meet 3.17.2 and 3.17.3 shall be provided with lifting and tiedown attachments conforming to 3.22 and 3.23, respectively.

3.19 Electromagnetic interference. The electromagnetic interference emission of the crusher shall conform the recommended limits specified in SAE J551 when measured in accordance with SAE J551.

3.20 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.20.1 Lubricants. The crusher shall operate as specified herein when lubricated with military lubricants.

3.20.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-354l, type I, II, or II. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Where oil holes are applicable, they shall have spring-loaded covers. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility. Grease fittings on electric motors shall be removed after being lubricated and replaced with threaded, metal plugs.

3.20.3 Pressure release device. An automatic pressure-release device shall be provided, where the use of pressure lubricating equipment could damage grease seals or other parts.

3.20.4 Enclosure with integral reservoir of lubricant. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, or sight glass or tube to determine the level of the lubricant. Each enclosure shall be equipped with a magnetic drain plug not less than 1/2-inch pipe size and with a means for filling the enclosure with lubricant. The drain plug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the crusher is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground. Accessibility to the drain plug, the filling means, and the lubricant-level checking device shall be obtained without the removal or adjustment of accessories or parts, except that panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed.

3.20.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. The crusher shall be run-in (broken-in) and delivered with these greases. The bearings shall be cleaned before lubrication because military greases may not be compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

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3.20.6 Oil lubrication. All components shall be filled to the operating level with the grades of oils conforming to MIL-L-2104 or MIL-L-2105 as required for the application, for use in the temperature range at destination. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.21 Fungus and moisture resistance. The electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with varnish conforming to MIL-V-173 except 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish shall be used in lieu of salicylanilide, the specified fungitoxicant, with the following exceptions:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component and element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish.

3.22 Lifting attachments. The crusher and sections (see 3.17.2 and 3.17.3) shall be provided with lifting attachments. The attachments shall enable the crusher and each section to be lifted in normal traveling or operating position. The attachments shall be located so that not less than 1-inch clearance is maintained between the slings and exterior parts of the crusher or section being lifted. Spreader bars are permitted. Each section shall be of sufficient strength to carry not less than 2-1/2 times the normal load imposed on the attachment. Attachments for the multiple slings shall be located so that the attached slings will converge not more than 19 feet above the lowest extremity of the load. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the attachment. The inside diameter of the eye shall not be less than 3 inches. Attachments shall be attached to members of sufficient strength to withstand stresses in the amount and direction of pull specified for the attachment. Where required, reinforcements shall be used to meet the requirements specified herein. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.23.

3.23 Tiedown attachments. The crusher and sections shall be provided with integral attachments to permit tiedown of the crusher or sections to the floor or deck of the transportation medium. The tiedown attachments shall withstand dynamic loads encountered during transport, and static loads of 8G forward, 8G rearward, 2-1/4-G upward, and 1-1/2-G sideward, where G equals the maximum shipping weight of the crusher or section. Each section and structural member to which the attachment is fastened shall withstand its proportion of the static load

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without permanent deformation or breakage. Attachments shall be beveled to prevent cutting of tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.22.

3.24 Identification marking. The crusher shall be identified in accordance with MIL-STD-130. The marking shall be applied to the crusher on identification plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. The location of lifting and tiedown attachments and recommended lifting and tiedown arrangements shall be included on these plates.

3.24.1 Instruction plates. Instruction plates shall be furnished and shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material, dimensions to suit the application. These plates shall indicate sequence for start up, shut down, and normal operation; production capacity, and all inherent dangers which can stem from unsafe operation or an unsafe operating position. All plates shall be securely attached to the crusher with machine screws, bolts and nuts, or rivets.

3.24.1.1 Warning plates. The following warning plates shall be furnished:

(a) A plate placed on the outside of the distribution panel (see 3.8.2.1.5), near main power inlet, with the following wording: "Warning. Install a ground rod and lead by bolting ground wire to grounding stud on crusher frame before this crusher is operated. A ground rod and lead should also be installed at power source (generator)."
(b) A plate placed adjacent to grounding stud or bolt (see 3.8.2.1.1) with the following wording: "Attach ground wire here."
(c) One plate located in a conspicuous place on the dolly having the following wording: "Warning. Do not connect this dolly to any semitrailer unless a 3-1/2-inch throat diameter kingpin is in place on the semitrailer."
(d) Warning plate specified in 3.13.12.

3.24.1.2 Wiring diagram. A plate containing a schematic operation wiring diagram of the high voltage system shall be mounted near the operator's position. The diagram shall show the physical location of all wiring interconnections in the same relationship as when they are installed. Identification marking of wires, instruments, control devices, and connections shall be shown and shall coincide with markings on the items installed. Low voltage vehicular wiring schematic shall not be included on this plate.

3.24.1.3 Production capacity plate. A production capacity plate similar to Figure 7 shall be provided. Capacity figures for the crusher shall be included on the plate.

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3.24.2 Shipping data plate. Shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class I material and in addition, shall show the silhouette of the crusher in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.25 Treatment and painting. The crusher, components, and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified.

3.26 Vehicle marking. The marking of the semitrailer shall be in accordance with MIL-STD-642. The GVW of the crusher and the weight of each sectionalized component shall be marked on the side in such a manner as to be discernible. The prescribed tire pressure shall be marked near the tires. Lifting and tiedown attachments shall be marked "LIFT HERE" or "TIEDOWN HERE", as applicable. Marking of lifting and tiedown attachments for sections shall include the word: "SECTIONALIZATION".

3.27 Technical publiclications to accompany the crusher. Such technical publications as are specified shall be furnished.

3.28 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.29 Toolbox. The crusher shall be provided with one or more weatherproof toolboxes made of steel. The toolbox lid shall be mounted with a piano-type hinge with a 0.125-inch-diameter brass pin. The toolbox lid shall open to at least 90 degrees from the closed position. The lid shall remain in the fully open position either by its own weight or by detent holding devices. Two tension-type fasteners with strikers shall be provided to retain the lid in the closed position. The fasteners and strikers shall be attached to the toolbox by either bolts or rivets. A hasp which will accommodate a lock shall be provided for locking the toolbox. The toolbox shall be fastened to the crusher in a protected position and the top of the box shall be not more than 4 feet from the ground. A drain hole with rubber plug shall be provided in the bottom of the toolbox. The toolbox shall be of a size to hold the emblem and blade (see 3.6), the ground rods (see 3.8.2.1.1), the jack pads (see 3.14.7), and the tools specified (see 3.28). The toolbox shall be marked to indicate the items it contains.

3.30 Items to be furnished with the crusher. No shipment of crushers shall be made unless repair parts, maintenance tools, and technical publications are included with the shipment or unless approval for shipment of the crushers without such items has been received from the contracting officer.

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3.31 Government-loaned property. When specified the following property in the quantities indicated will be loaned by the Government (see 6.5):

Item No.	Description	Identification	Quantity
1	M916 – Truck Tractor, 20-ton, 6 x 6	(CCE)	1
2	M920 – Truck Tractor, 20-ton, 8 x 6	(CCE)	1
3	M917 – Truck, Dump, 20-ton, 8 x 6	(CCE)	1

3.32 Workmanship. The workmanship of each crusher shall be in accordance with the engineering, manufacturing, and production standards of the construction equipment industry. Workmanship deficiencies if any, shall be discovered and corrected. All parts, components, and assemblies of the crusher including bearings, seals, machinery, stampings, welded parts, casting, forgings, and machined work shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material. External surfaces shall be free of burrs and sharp edges and corners except when sharp edges and corners are functional.

3.32.1 Steel fabrication. Steel used in the fabrication of the crusher shall provide original quality surface finish and shall be free from kinks and sharp bends. Steel having an eroded surface is not acceptable. The forming of the material shall be done by methods that will not cause damage to the metal. Shearing, punching, and chipping shall be done uniformly, neatly, and accurately. Corners shall be square and true, and all sharp edges and burrs shall be removed. The burned surfaces of flame-cut material shall be free of burrs, slag, and sharp edges. Precautions shall be taken to avoid overheating of metal. Heated metal shall be allowed to cool slowly, except where heat treatment is required. All bends of a major character shall be made with precise, unyielding dies or jig fixtures to insure uniformity of size and shape.

3.32.2 Welders and welding.

3.32.2.1 Welders. Before assinging any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by any of the following listed codes or an equivalent manufactures qualification test for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualification of the ASME Boiler and Pressure Vessel Code.

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3.32.2.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are tested as specified herein.

3.32.3 Bolted connections. Boltholes shall be accurately punched or drilled and shall have burrs removed. Washers or lockwashers shall be provided. All bolts, screws, and nuts shall be tight and correctly torqued. Unless otherwise specified herein, all bolts and screws (except set screws and where lock nuts are used) shall be provided with spring-type lockwashers. Fasteners shall be lubricated prior to wrenching and installation. Torque tools shall be proofed at set intervals to insure constant and correct torque performance.

3.32.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.32.5 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for its intended use.

3.32.6 Forgings. Forged components shall be in accordance with the Forging Industry Handbook of the Forging Industry Association.

3.32.7 Machine work. All machined parts shall be manufactured to gage through the use of correct jigs and fixtures and shall permit interchangeability of these parts.

3.32.8 Human factors. The crusher shall be designed to be operable and maintainable by the 5th and 95th percentile operator, as listed in SAE J833, suitably dressed for the environment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The crusher shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The crusher shall be tested as specified in 4.6.1 and 4.6.2 through 4.6.2.5. Failure to pass any test shall be cause for rejection.

4.4 Initial production inspection. When an initial production crusher is specified (see 3.3), crushers will be selected at random by the Government from the first five crushers being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 through 4.6.2.3.7 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production crusher shall not exclude the remaining crushers from the quality conformance inspection and acceptance provisions specified in Section 4.

4.4.1 Inspection failure. Failure of an initial production crusher to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for refusal by the Government to continue acceptance of crushers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on crushers previously accepted and produced under the contract any deficiencies found as a result of the initial production testing will be considered prima facie evidence that all crushers accepted prior to the completion of initial production testing are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each crusher shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each crusher shall be tested as specified in 4.6.3.5. Failure to pass test shall be cause for rejection.

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4.6 Inspection procedure.

4.6.1 Examination. The crusher shall be examined for the defects listed in table 1.

TABLE I. Examination schedule (Requirements).

Number	Characteristics	Requirement Paragraph
Major		
101.	Dimensions that affect form, fit, or function not as specified.	3.1
102.	Ease of maintenance not as specified.	3.1.2
103.	Material not as specified.	3.4
104.	Safety not as specified.	3.6
105.	Hoppers and chutes not as specified.	3.7
106.	Diesel engine not as specified.	3.8.1
107.	Air cleaner not as specified.	3.8.1.1
108.	Clutch and clutch controls not as specified.	3.8.1.2
109.	Fuel priming pump not as specified.	3.8.1.3
110.	Rain cap not as specified.	3.8.1.4
111.	Electric motors not as specified.	3.8.2
112.	Power cable, reels, and ground rods not as specified.	3.8.2.1.1
113.	Power cable adaptor assembly not as specified.	3.8.2.1.1.1
114.	Auxiliary branch circuits not as specified.	3.8.2.1.2.1
115.	Main feeder receptacle not as specified.	3.8.2.1.2.2
116.	Auxiliary main feeder outlet not as specified.	3.8.2.1.3
117.	Safety cutouts not as specified.	3.8.2.1.4
118.	Distribution panel and branch circuit protective devices not as specified.	3.8.2.1.5
119.	Motor starters not as specified.	3.8.2.1.6
120.	Internal wiring not as specified.	3.8.2.1.7
121.	Motor mounting brackets not as specified.	3.8.2.2
122.	Chain drivers not as specified.	3.9.1
123.	V-belt drivers not as specified.	3.9.2
124.	Gear drives not as specified.	3.9.3
125.	Shafting not as specified.	3.9.4
126.	Keys and keyways not as specified.	3.9.5
127.	Torque arm speed reducers not as specified.	3.9.6
128.	Feeder not as specified.	3.10
129.	Grizzly not as specified.	3.11
130.	Screen activating mechanism not as specified.	3.11.1
131.	Shaft bearings not as specified.	3.11.1.1
132.	Woven wire screen cloth not as specified.	3.11.2
133.	Material for the screen cloth not as specified.	3.11.2.1
134.	Wire size not as specified.	3.11.2.2
135.	Weave of screen cloth not as specified.	3.11.2.3

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TABLE II. Examination schedule (Requirements) (Cont'd).

Number	Characteristic	Requirement Paragraph
136.	Crushing component not as specified.	3.12
137.	Frame for crushing component not as specified.	3.12.1
138.	Jaw and cheek plates not as specified.	3.12.2
139.	Pitman not as specified.	3.12.3
140.	Pitman shaft bearings not as specified.	3.12.4
141.	Pitman shaft not as specified.	3.12.5
142.	Flywheels not as specified.	3.12.6
143.	Toggle plate not as specified.	3.12.7
144.	Adjusting mechanism not as specified.	3.12.8
145.	Jaw setting not as specified.	3.12.9
146.	Conveyors not as specified.	3.13
147.	Conveyor frame not as specified.	3.13.1
148.	Head pulley not as specified.	3.13.2
149.	Tail pulley not as specified.	3.13.3
150.	Taper-type locking device (pulley and shaft) not as specified.	3.13.4
151.	Bearing for the head and tail shafts not as specified.	3.13.5
152.	Takeup adjustment not as specified.	3.13.6
153.	Belt carriers not as specified.	3.13.7
154.	Troughing rolls not as specified.	3.13.7.1
155.	Impact rolls not as specified.	3.13.7.2
156.	Return rolls not as specified.	3.13.7.3
157.	Discharge conveyor belt not as specified.	3.13.8
158.	Belt fasteners not as specified.	3.13.8.1
159.	Driving mechanism not as specified.	3.13.9
160.	Holdback device not as specified.	3.13.9.1
161.	Belt cleaners not as specified.	3.13.10
162.	Receiving hopper not as specified.	3.13.11
163.	Side delivery conveyor not as specified.	3.13.12
164.	Trailer components not as specified.	3.14
165.	Trailer frame not as specified.	3.14.1
166.	Upper fifth wheel and kingpin not as specified.	3.14.1.1
167.	Rear towing components not as specified.	3.14.1.2
168.	Front and rear intervehicular connection not as specified.	3.14.1.3
169.	Rear suspension not as specified.	3.14.2
170.	Wheel assemblies not as specified.	3.14.3
171.	Chock blocks not as specified.	3.14.4
172.	Brake system not as specified.	3.14.5
173.	Brake lines not as specified.	3.14.5.1
174.	Vehicular electric system not as specified.	3.14.6
175.	Stabilizing jacks not as specified.	3.14.7

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TABLE I. Examination schedule (Requirements) (Cont'd).

Number	Characteristic	Requirement Paragraph
176.	The dolly furnished by the contractor will contain the following items: Fifth wheel, drawbar coupler, landing leg, dolly systension, dolly electrical system, dolly intervehicular connections and safety chains.	3.15, 3.15.1.1, 3.15.2, 3.15.3, 3.15.4, 3.15.5, 3.15.6, 3.15.7
177.	Operator's platform not as specified.	3.16
178.	Auxiliary components for platform not as specified.	3.16.1
179.	Controls for the platform not as specified.	3.16.2
180.	Lifting and tiedown attachments not as specified.	3.18, 3.22, 3.23
181.	Means for lubrication not provided as specified.	3.20
182.	Lubricants not as specified.	3.20.1
183.	Lubrication fittings not as specified.	3.20.2
184.	Pressure release device not as specified.	3.20.3
185.	Enclosure with integral reservoir of lubricant not as specified.	3.20.4
186.	Grease lubrication not as specified.	3.20.5
187.	Oil lubrication not as specified.	3.20.6
188.	Fungus and moisture resistance not as specified.	3.21
189.	Instruction plates not as specified.	3.24.1
190.	Warning plates not as specified.	3.24.1.1
191.	Wiring diagram not as specified.	3.24.1.2
192.	Production capacity plate not as specified.	3.24.1.3
193.	Shipping data plate not as specified.	3.24.2
194.	Vehicle marking not as specified.	3.26
195.	Steel fabrication not as specified.	3.32.1
193.	Welders not certified as specified.	3.32.2.1
197.	Welding not as specified.	3.32.2.2
198.	Bolted connections not as specified.	3.32.3
199.	Riveted connections not as specified.	3.32.4
301.	Castings not as specified.	3.32.5
302.	Forgings not as specified.	3.32.6
303.	Machine work not as specified.	3.32.7
Minor		
201.	Identification marking not as specified.	3.24
202.	Treatment and painting not as specified.	3.25
203.	Technical publications not furnished as specified.	3.27

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TABLE I. Examination schedule (Requirements) (Cont'd).

Number	Characteristic	Requirement Paragraph
204.	Repair parts and maintenance tools not furnished as specified.	3.28
205.	Toolbox not as specified.	3.29
206.	Workmanship not as specified.	3.32

4.6.2 Tests. Prior to test, the crusher shall be lubricated with military oils and greases specified in 3.20. Oils shall be those designated for use in the ambient temperature at which the test is being performed. The contractor shall furnish equipment to feed crusher and portable conveyors to carry away crushed material during test.

4.6.2.1 Test conditions (operating). The crusher shall be tested either as a unit as specified in 3.1.1, or as a plant with a secondary crusher as specified in 3.1.1. Quarry rock shall be of the material as specified in 3.1.1.1 containing some rock 18 inches by 34 inches. Only that maintenance established by the contractor and submitted as a maintenance schedule prior to the tests shall be performed during the test.

4.6.2.1.1 Percentage of crush. Arrange the flow of material so that all material passing through the vibrating grizzly bars will be discharged to the stub conveyor or chute. Weigh not less than 100 tons of crushed material discharged from the delivery conveyor. The material bypassed through the grizzly bars shall be weighed separately and should not exceed 20 percent of the combined total of material from the discharge chute and the delivery conveyor. Presence in the feed of more than 20 percent of material not requiring crushing shall constitute failure of this test.

4.6.2.1.2 Operating time. Crushed rock production capacity shall be computed on the basis of net operating time which shall be defined as gross operating time less allowable down time for the following:

(a) Jamming from rock fragments larger than the maximum size of rock specified in 3.12.
(b) Delay due to lack of material delivery to the crusher.
(c) Normal machinery maintenance (lubrication, jaw setting adjustments, takeups) (down time due to any design deficiency shall not be considered normal maintenance).
(d) Other conditions considered by the contracting officer to be beyond the control of the contractor.

Down time required for maintenance other than lubrication and operating adjustments shall be considered as net operating time. Down time caused by design deficiencies or failure of parts not classified as high mortality parts shall constitute failure of any test.

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4.6.2.1.3 Jaw settings. When the crusher is tested as a unit, the jaw settings shall be as follows:

Closed setting in inches	Time in hours
2	250
Maximum (not less than 5 inches)	10
Minimum (not more than 2 inches)	10

When the crusher is tested as a plant, the jaw settings shall be compatible with the roll crusher to produce the following sizes:

Plant product size	Time in hours
1-1/2 inch minus	250
3/4 inch minus	10
2-1/2 inch minus	10

Capacity in tons per hour and corresponding jaw setting shall be recorded as part of test reports.

4.6.2.2 Capacity as a plant.

4.6.2.2.1 Capacity as a unit. Operate the crusher under the applicable test conditions specified in 4.6.2.1 through 4.6.2.1.3. Cumulative daily runs are permissible. Nonconformance to test conditions or to 3.1.1(a) and 3.12 shall constitute failure of this test.

4.6.2.2.2 Capacity as a plant. Operate the crusher under the applicable test conditions specified in 4.6.2.1 through 4.6.2.1.3. Cumulative daily runs are permissible. Nonconformance to test conditions or to 3.1.1(b) and 3.12 shall constitute failure of this test.

4.6.2.2.3 Flywheel test. During the capacity tests, measure the speed of the flywheel for a continuous period of 3 minutes, during which the crusher is loaded to maximum capacity with rock. Compare this flywheel speed to the flywheel speed when the crusher is operating empty. Nonconformance to 3.12.6 shall constitute failure of this test.

4.6.2.2.4 Uncrushable material test. With the crusher operating at capacity rate, drop a piece of uncrushable material such as tramp iron, into the crusher. The piece shall be larger than the maximum jaw opening at the jaw setting being used, but not more than 2 inches larger. Nonconformance to 3.12.7 shall constitute failure of this test.

4.6.2.2.5 Apron feeder test. With the loading hopper empty and the feeder operating, discharge one rock with dimensions not greater than maximum crusher size and weighing not less than 1,000 pounds onto the apron feeder from a height of 8 feet above the feeder surface. Repeat this test not less than 10 times. Any operating difficulty, permanent deformation, or breakage of any part of the crusher shall constitute failure of this test.

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4.6.2.2.6 Holdback device test. During capacity and performance tests cut off electric power to motor-driven conveyors when carrying capacity load to determine conformance to 3.13.9.l. Failure to prevent slippage or reversal shall constitute failure of this test.

4.6.2.3 Transportability tests.

4.6.2.3.1 Highway and cross-country tests. Tow the crusher as follows:

(a) Before any highway and cross-country tests are conducted, measure all critical mounting points on the crusher. After all highway and cross-country tests are completed, check the measurement of critical mounting points and inspect all welds.
(b) 250 miles over highways with the M916 truck-tractor at speeds up to 30 mph.
(c) 10 miles cross-country with the dolly specified in 3.15 and the M917 dump truck ballasted as required to provide traction.
(d) 10 miles cross-country with the M920 truck-tractor at speeds up to 10 mph.
(e) Tow over 8-inch high obstacles such one pass perpendicular and one pass at an angle 45 degrees from both right and left that all left and right combinations of wheels have simultaneously passed over the obstacles. Observe tire-to-surface contact, evidence of binding, or interference. After returning the crusher to level ground, measure the diagonals and parallels for evidence of axle misalinement.
(f) Tow over V-ditches not less than 36 inches wide and 18 inches deep. Negotiate the ditches two passes each from both perpendicular and at an angle of 45 degrees to the direction of travel.
(g) Ford the semitrailer through a body of water not less than 30 inches deep. While in the water turn off the lights for approximately 1 minute and then return to "on".
(h) Measure and record the deviation or off-tracking of the semi-trailer with respect to the dolly and to the prime movers when the semi-trailer is towed straight ahead on a paved surface.
(i) Demonstrate brake system conformance to DOT regulations.
(j) With the truck-tractor only, tow the semitrailer up to 30 mph and apply the brakes to produce a deceleration rate of not less than 10 feet per second, until slowed to 10 mph. Immediately accelerate to 30 mph. Repeat this snubbing cycle for a total of 5 cycles and make a panic stop on the 6th cycle.
(k) At 10 mph, cause the crusher to go into "emergency", simulating a breakaway from the dump truck and dolly-semitrailer.
(l) Test the brake system in accordance with TTMA recommended practice RP#12.

During the above tests, lights shall be on and turn signals actuated on turns. Upon completion of the travel, remove the rims and wheels and examine the entire crusher, including the rims and wheels. Damage to or permanent deformation of any part of the crusher, change in critical mounting point dimensions, evidence of any weld being defective, inability to complete any portion of the test, incompatibility with any towing vehicle, or operating difficulty shall constitute failure of this test.

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4.6.2.3.2 Truck and dolly-semitrailer compatibility test. With the Government-loaned M917 dump truck and dolly-semitrailer in towing position, jackknife the truck-dolly-semitrailer combination until the centerline of the truck is at an angle of at least 60 degrees with the centerline of the dolly. Next, tow the dolly-semitrailer combination at minimum truck turning radius to the right and to the left. Test all lights. Repeat the tests using the Government-loaned M916 truck-tractor in lieu of the M917 dump truck. Interference between the truck and any part of the dolly or semitrailer, or lights not functioning shall constitute failure of this test.

4.6.2.3.3 Truck and semitrailer compatibility test. Couple the Government loaned M916 truck-tractor to the semitrailer and tow the combination at the minimum turning radius of the truck to the right and to the left. Repeat the tests with the Government-loaned M920 truck-tractor. Jackknife each truck-tractor to an angle of at least 90 degrees with respect to the semitrailer. Incompatibility with either the truck or truck-tractor, or interference with the truck or truck-tractor by any part of the trailer shall constitute failure of this test.

4.6.2.3.4 Dolly and semitrailer compatibility test. With the dolly and semitrailer in travel position, hold the lunette between 32 and 34 inches from the ground and turn the towbar at least 110 degrees to the right and to the left of the centerline of the semitrailer. Interference between the dolly and any part of the semitrailer shall constitute failure of this test.

4.6.2.3.5 Railroad and air transport. Sectionalize and reassemble the crusher. Nonconformance to 3.17.2, and 3.17.3 shall constitute failure of these tests. The tests for sectionalization and for lifting and tiedown attachments may be accomplished in conjunction with the tests specified in 4.6.2.3.6.

4.6.2.3.6 Lifting and tiedown attachments. Apply the static load specified in 3.22 for lifting attachments and 3.23 for tiedown attachments after the attachments are mounted on the crusher or section, as applicable. The direction of the applied load shall be in the direction in which the load is normally applied. Weld failure or permanent deformation of the attachment or of any component of the crusher or section shall constitute failure of this test.

4.6.2.3.7 Tiedown attachments (dynamic). Anchor each section of the crusher independently, as sectionalized for rail transport, to a flatcar using only the tiedown attachments. Use no blocking. Hump the flatcar at a speed of 7 miles per hour against five boxcars with their brakes set. Repeat the test with the opposite end of the section in the direction of the five cars. Any permanent deformation or breakage of the tiedown attachments or parts of the section shall constitute failure of this test.

4.6.2.4 Electromagnetic interference. The preproduction model shall be tested to determine conformance with 3.19. The contractor shall furnish the contracting officer the report of tests. Disapproval of the report shall constitute failure of this test.

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4.6.2.5 Production test. Operate the crusher for a period of 1 hour without processing material. Test the brake system in accordance with TTMA RP#12. Any operational difficulties, breakage, or permanent deformation shall constitute failure of this test.

4.7 Inspection comparison. The Government may select crushers at any time during the contract production period and subject these crushers to the examination specified in 4.6.1 and the tests specified in 4.6.2 through 4.6.2.3.7 to determine that the quality of the selected crusher is equal to the quality standards established during initial production testing. The inspection will be performed by the Government at a site selected by the Government. Crushers will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production crushers.

4.7.1 Inspection failure. Failure of any crusher when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of crushers until objective evidence furnished by the contractor reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Test. The preproduction pack of crated components for Level A shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for characteristics listed in table II the following defects. AQL shall be 2.5 percent defective, for major defects and 4.0 percent defective for minor defects.

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TABLE II. Examination schedule (Packaging).

Numbers	Characteristics	Requirement Paragraph
Major		
304.	Preproduction pack not as specified	5.1
305.	Preservation not level A, level B, or commercial as specified (see 6.2).	5.2
306.	Crusher is not sectionalized for protection and level A as specified.	5.2.1.1
307.	Matchmarking for level A not as specified.	5.2.1.2
308.	Preservatives for level A not as specified.	5.2.1.3
309.	Unfinished surfaces not protected and preserved for level A as specified.	5.2.1.4.1
310.	Machined surfaces not preserved for Level A as specified.	5.2.1.4.2
311.	Components of the semitrailer and dolly not preserved for level A as specified.	5.2.1.5
312.	Engine and engine accessories not preserved for level A as specified.	5.2.1.6
313.	Clutch not preserved for level A as specified.	5.2.1.7
314.	Instruments not protected or preserved for level A as specified.	5.2.1.8
315.	Electric motors not preserved for level A as specified.	5.2.1.9
316.	Openings on electrical components not sealed with tape as specified.	5.2.1.10.1
317.	Power cable and reels not preserved for level A as specified.	5.2.1.10.2
318.	Ground rod assemblies, clamps, and wires not placed in container for level A as specified.	5.2.1.10.3
319.	Power cable adapter assembly not placed in a close-fitting box for level A as specified.	5.2.1.10.4
320.	Exposed gears not preserved for level A as specified.	5.2.1.11.1
321.	Enclosed gears not preserved for level A as specified.	5.2.1.11.2
322.	Exposed chain not preserved for level A as specified.	5.2.1.12.1
323.	Enclosed chains not preserved for level A as specified.	5.2.1.12.2
324.	Shafts and universal joints not preserved for level A as specified.	5.2.1.13
325.	Drive belts and pulleys not preserved for level A as specified.	5.2.1.14
326.	Conveyor belts and fasteners not prepared for level as specified.	5.2.1.15

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Table II. Examination schedule (Packaging) (Cont'd).

Number	Characteristic	Requirement Paragraph
327.	Loose components not protected by boxing for for level A as specified.	5.2.1.19
328.	Crusher not preserved for level B as specified.	5.2.2
329.	Semitrailer components not preserved for level B as specified.	5.2.2.1
330.	Clutch not preserved for level B as specified.	5.2.2.2
331.	Electrical components openings for level B not sealed with tape as specified.	5.2.2.3
332.	Exposed chains not preserved for level B as specified.	5.2.2.4
333.	Drive belts not released from tension for level B as specified.	5.2.2.5
334.	Conveyor belt fasteners are coated with varnish for level B not as specified.	5.2.2.6
335.	Consolidation for level B not as specified.	5.2.2.7
336.	Crusher not preserved for Commercial as specified.	5.2.3
337.	Packaging not level A or Commercial as specified.	5.3
338.	Packing for level A not as specified.	5.3.1
339.	Packing for Commercial not as specified.	5.3.2
Minor		
207.	Technical publications not preserved for level A as specified.	5.2.1.16
208.	Maintenance tools not preserved for level A as specified.	5.2.1.17
209.	Repair parts not preserved for level A as specified.	5.2.1.18
210.	Depreservation guide not furnished as specified.	5.4
211.	Marking not in accordance with MIL-STD-l29.	5.5

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2), to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the preproduction model or a production model. When the preproduction model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model and the production model.

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5.2 Preservation. Preservation shall be level A, level B, or level C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Sectionalization and disassembly. The crusher shall be sectionalized to meet the applicable requirements of 3.17.2 or 3.17.3, as applicable. In addition, components of the crusher and sections vulnerable to damage, pilferage, and loss, and protruding parts that increase cubage and that can be removed and reassembled without special tools or skills shall be removed. Clusters of lights shall be removed and placed in the toolbox. Bolts, nuts, pins, and washers removed shall be replaced in one of the mating parts and secured to prevent their loss. Keys removed shall be secured in the keyway of the pulley or flywheel.

5.2.1.2 Matchmarking. To facilitate reassembly, components and sections removed shall be matchmarked. Large sections and components shall be matchmarked identically by stencilled letters or numerals with lusterless white enamel, overcoated with varnish conforming to TT-V-121 or identically matchmarked with paint conforming to MIL-P-13983. Small components removed shall have matchmarking information on cloth tags conforming to UU-T-81 or on metal tags. The information on the tags shall be applied with waterproof ink. The tags shall be securely attached to the components so as to prevent damage to the preservation or to the components.

5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unprotected surfaces.

5.2.1.4.1 Unfinished surfaces. Unpainted exterior metal surfaces of components of the crusher such as hoppers, chutes, feeders, crusher frame, jaw and cheek plates, pitman, flywhee1s, toggle plates, adjusting mechanisms, rods, screws, nuts, bolts, clevises, nonprecision chains, conveyor rolls, sprockets, pulleys, ratchets, linkages, chock blocks, safety chains, coupling pins, towing hooks, and any other surface exposed by sectionalization or disassembly, and any other similar unpainted exterior metal surface not definitely specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-l preservative.

5.2.1.4.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with type P-6 or P-11 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2 and secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.5 Semitrailer and dolly components. The components of the semitrailer and dolly shall be preserved in accordance with the level A (mobile) preservation and packaging requirements of MIL-STD-281.

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5.2.1.6 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, level A, type I, method I, using the water and antifreeze procedure for the cooling system except the antifreeze shall conform to MIL-A-46153.

5.2.1.7 Clutch. With the cover plate removed and with the clutch engaged, all interior components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664. Coating of nonmetallic surfaces shall be kept to a minimum. Clutch control mechanisms not enclosed shall be coated with Type P-l preservative. The clutch shall be completely disengaged.

5.2.1.8 Instruments. Instruments not protected with a cover or housing shall have the dial glasses of instruments or the complete panel covered with a fitted piece of 1/4 inch thick plywood or hardboard and secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.9 Electric motors. Electric motors shall be preserved in accordance with MIL-E-16298, level A, using the alternate method as specified for assembled machines.

5.2.1.10 Electrical components.

5.2.1.10.1 Openings. Openings into the starter switches, controls, junction boxes, power receptacles, lamps, and ends of the intervehicular cable and main feeder cables shall be sealed with tape conforming to PPP-T-60, type IV.

5.2.1.10.2 Power cables and reels. Unpainted exterior surfaces of the cable reels shall be coated with type P-1 preservative. The power cables shall be evenly wound on their reels and secured to prevent unwinding.

5.2.1.10.3 Ground-rod assemblies, clamps, and wires. The ground-rod assemblies, clamps, and wires shall be placed in the container provided.

5.2.1.10.4 Power cable adaptor assembly. The power cable adaptor assembly shall be placed in a close-fitting box conforming to PPP-B-636, W6c.

5.2.1.11 Gears.

5.2.1.11.1 Exposed gears. Exposed gears shall be coated with type P-l preservative or with primer conforming to TT-P-664.

5.2.1.11.2 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30, or 50) shall have the housing filled to the operating level with type P-10, type I, grade as applicable and tagged to indicate: "The preservative is good for operation. Do not drain until the first required lubrication change." Gears operating on gear lubricant (SAE 80 or 90) shall be filled to the. operating level with lubricant conforming to MIL-L-2105, grade as applicable, and tagged

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to indicate: "The lubricant is good for operation." Gears not operating on lubricating oil (SAE 10, 30, or 50) or lubricant (SAE 80 or 90) shall be filled to the operating level with the approved lubricant required for operation. The gear housings shall be tagged to indicate: "The housing is filled to the operating level with the applicable type lubricant." The information shall be on metal tags or cloth tags conforming to UU-T-8l. The information on the tags shall be applied with waterproof ink.

5.2.1.12 Drive chains.

5.2.1.12.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. The excess preservative shall be drained, and the entire chain shall be coated with type P-l preservative. Removed chains shall be coiled, and the coil shall be secured with soft, annealed wire and wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured with tape conforming to PPP-T-60, type IV. Sprockets and other unprotected surfaces of the drives shall be coated with type P-l preservative.

5.2.1.12.2 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified in 5.2.1.11.2 for enclosed gears and gear housings.

5.2.1.13 Shafts and universal joints. Exposed surfaces of shafts and universal joints shall be coated with type P-l preservative.

5.2.1.14 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved in accordance with MIL-P-116, Method IC-l or IC-3. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664.

5.2.1.15 Conveyor belts and fasteners. Metal belt fasteners shall be coated with varnish conforming to TT-V-121. The belt shall be released from tension. When the conveyor is sectionalized, the belt shall be removed and compactly rolled, and the roll shall be secured with strapping conforming to QQ-S-781, class I, type l or IV, size as applicable, finish B. The straps shall be placed circumferentially on the roll and the straps tensioned so as not to damage the belt.

5.2.1.16 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-l or IC-3 and placed in the toolbox.

5.2.1.17 Maintenance tools. Maintenance tools shall be preserved in accordance with the Level A preservation and packaging requirements of PPP-P-40 and placed in the toolbox.

5.2.1.18 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts. The repair parts shall be placed in the toolbox when space is available or consolidated in boxes as specified in 5.2.1.19.

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5.2.1.19 Consolidation. Loose components requiring the protection of a container shall be consolidated in close-fitting boxes conforming to PPP-B-60l, Overseas Type, style optional, or PPP-B-62l, class 2, style optional. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall not be required.

5.2.2 Level B. The complete crusher shall be preserved as specified for level A, with the following exceptions.

5.2.2.1 Semitrailer components. The components of the semitrailer and dolly shall be preserved and packaged in accordance with MIL-STD-281, level B (mobile).

5.2.2.2 Clutch. The clutch shall be completely disengaged. Coating of interior costs of the clutch with primer shall not be required.

5.2.2.3 Electrical component openings. Openings into electrical components specified in 5.2.1.10.1 shall not require sealing with tape.

5.2.2.4 Exposed chains. Exposed drive chains shall be preserved as specified for level A, except only type P-3 or P-9 preservative shall be applied.

5.2.2.5 Drive belts. Drive belts shall be released from tension.

5.2.2.6 Conveyor belt fasteners. Belt fasteners are not required to be coated with varnish.

5.2.2.7 Consolidation. Consolidation shall be as specified for level A, except boxes shall be Domestic type or class as applicable.

5.2.3 Level C. Each complete crusher with repair parts, tools, and publications shall be preserved in accordance with MIL-STD-1188 and the following specific requirements. The engine crankcase and all lubricant reservoirs shall be filled to operating level with the lubricant recommended for operation. The cooling system shall be filled with a fresh, clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153.

5.3 Packing. Packing shall be level A or C as specified.

5.3.1 Level A. The basic crusher shall be prepared for mobile shipment. The toolbox lid shall be closed and secured to prevent pilferage. When sufficient space is available, components and sections shall be placed and secured on the crusher in any available space that will not interfere with lifting or mobility. The components shall be secured to the crusher with zinc-coated strapping conforming to QQ-S-78l. Components and sections that cannot be secured to the crusher shall be packed in crates conforming to MIL-C-52950 or MIL-C-3774, nailed assembly, skid-type base. Blocking, bracing, and anchoring shall be in accordance with the appendix to the applicable crate specification and with MIL-STD-1186. Crated components shall be shipped at the same time as the mobile basic crusher.

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5.3.2 Level C. The complete crusher shall be prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest rating. The basic crusher shall be prepared for mobile shipment. The toolbox lid shall be closed and secured to prevent pilferage. When sufficient space is available, components and sections shall be placed and secured on the crusher in any available space that will not interfere with lifting or mobility. Components and sections that cannot be secured to the crusher shall be packed in crates complying with Uniform Freight Classification rules or National Motor Freight Classification rules. Crated components shall be shipped at the same time as the mobile basic crusher.

5.4 Depreservation guide. Two depreservation guides shall be prepared. One guide shall be placed in a waterproof envelope marked "Depreservation Guide" and secured in a conspicuous location on the crusher. The other guide shall be placed in the package with the technical manuals and the package marked to indicate the depreservation guide is inside. Unless otherwise specified (see 6.2), DA Form 2258 "Depreservation Guide for Vehicles and Equipment" shall be used.

5.5 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The jaw crusher is intended for use as the primary rock-crushing unit in a rock crushing and screening plant (75 TPH).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2).
(c) When an initial production crusher is required and number of crushers required for initial production inspection (see 3.3).
(d) When 24-volt service lighting system is required.
(e) When jacks will be furnished by the Government (see 3.14.7).
(f) Color required (see 3.25).
(g) Technical publications required (see 3.27).
(h) Applicable list of repair parts and maintenance tools (see 3.28).
(i) Whether the crusher shall be tested as a unit or as a plant (see 4.6.3.1).
(j) Levels of preservation and packaging and levels of packing required (see 5.2 and 5.3). Level B preservation and packaging is intended.
(k) Time frame required for submission of the preproduction pack (see 5.1).
(l) Degree of preservation and packing required (see 5.2 and 5.3).
(m) When other than DA Form 2258 is to be used (see 5.4).

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6.3 Preproduction model. Any changes or deviations of production crushers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish crushers conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the crusher in accordance with this specification.

6.5 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.31.

6.6 DA Form 2258. The contracting officer shall arrange to furnish DA Form 2258 when requested by the contractor.

6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodians:
Army – ME

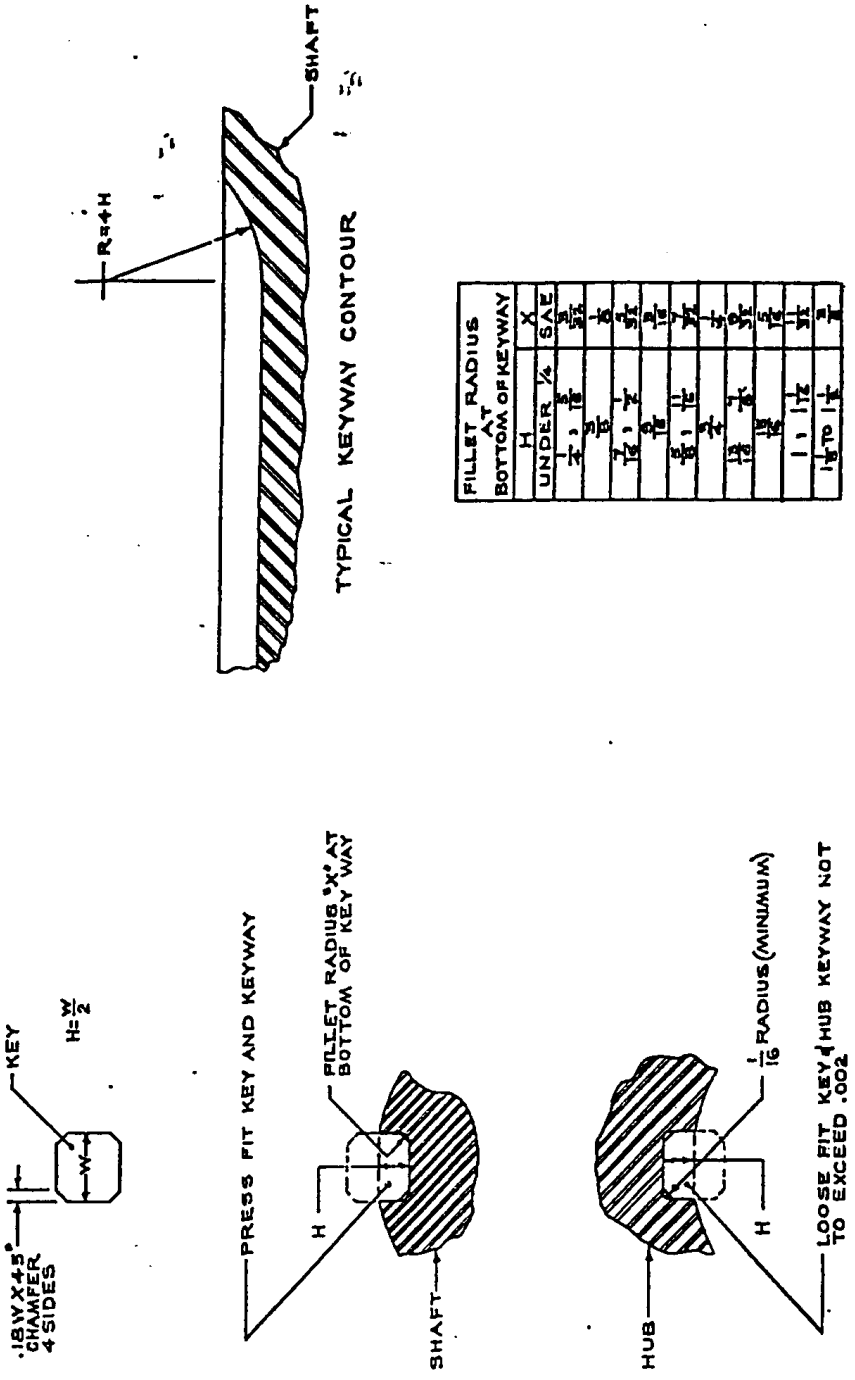
Preparing activity:
Army - ME

Review activity:
Army - AT
DLA - CS

Project 3820-0166

User activities:
Army - CE
Navy – MC

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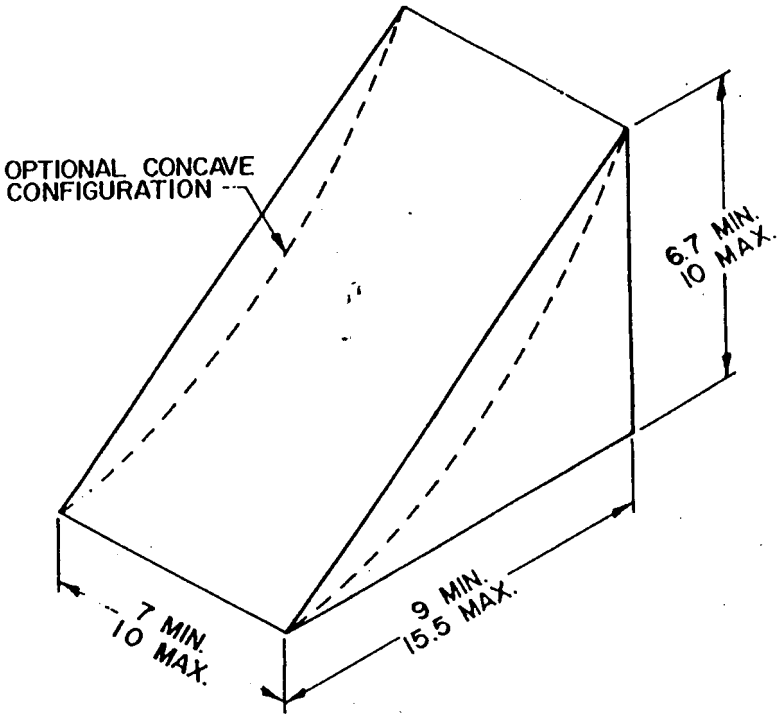


FILLET RADIUS AT BOTTOM OF KEYWAY		
H		X
OVER	TO	SAE
1/4	3/8	1/16
3/8	7/16	3/32
7/16	9/16	1/8
9/16	7/8	5/32
7/8	1 1/4	3/16
1 1/4	1 3/8	1/4
1 3/8	1 3/4	5/16
1 3/4	2 1/4	3/8

FIGURE 1. KEYS AND KEYWAYS

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CHOCK MATERIAL NOTE

1. ALUMINIUM
2. MAGNESIUM
3. SEMI-RIGID ELASTOMER
4. MAXIMUM CHOCK WEIGHT-11.5 LB.

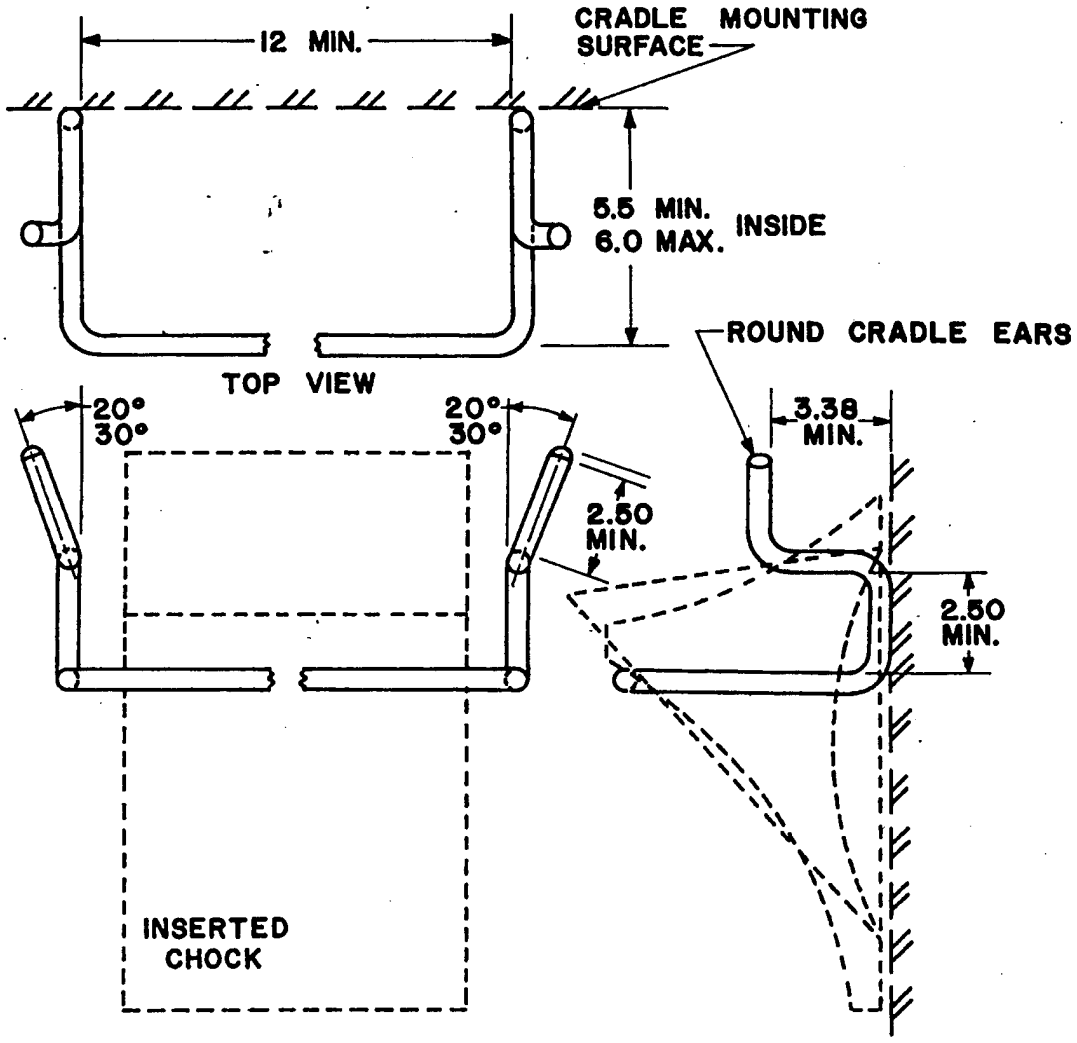
CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

FIGURE 2. CHOCK BLOCK ASSEMBLY.

X 603 A

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CRADLE MATERIAL NOTE: MINIMUM 1/2-INCH DIA. STEEL ROD. BEND RADII - 1.0-IN. MIN.

FIGURE 3. CRADLE

X-604A

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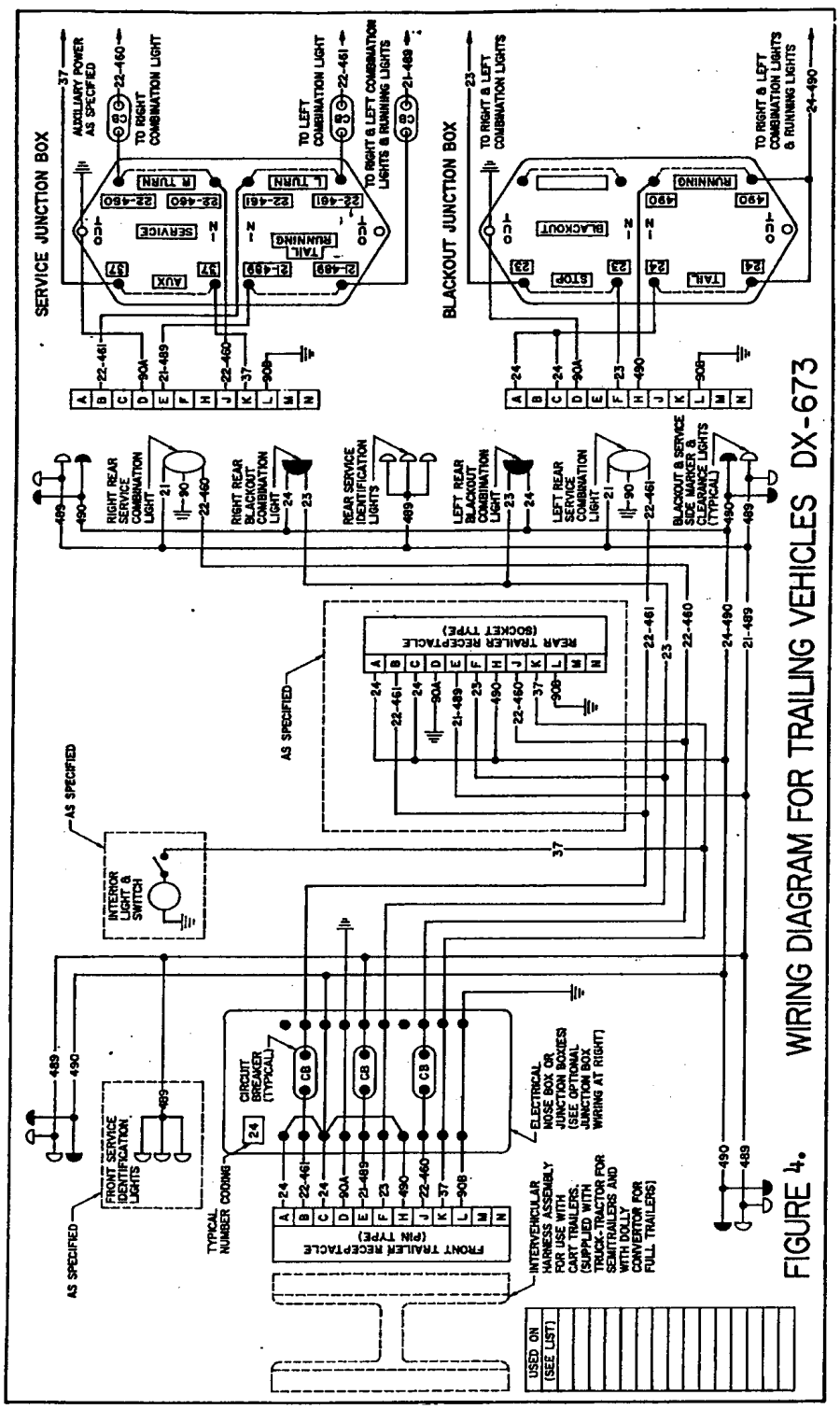
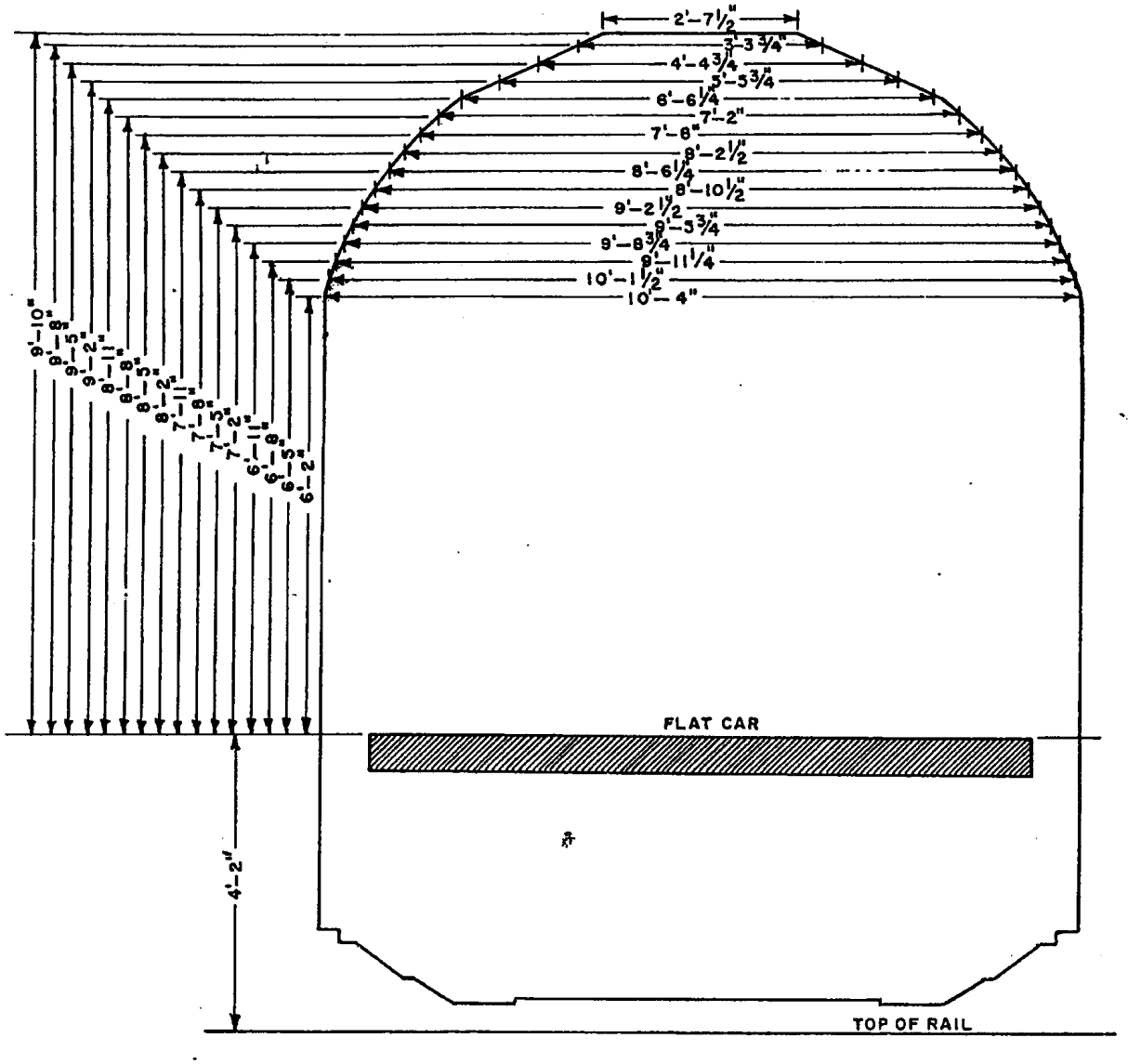


FIGURE 4. WIRING DIAGRAM FOR TRAILING VEHICLES DX-673

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ITEM LENGTH-32'-10"

Figure 5. BERNE INTERNATIONAL
(CLEARANCE DIAGRAM)

A13123-1

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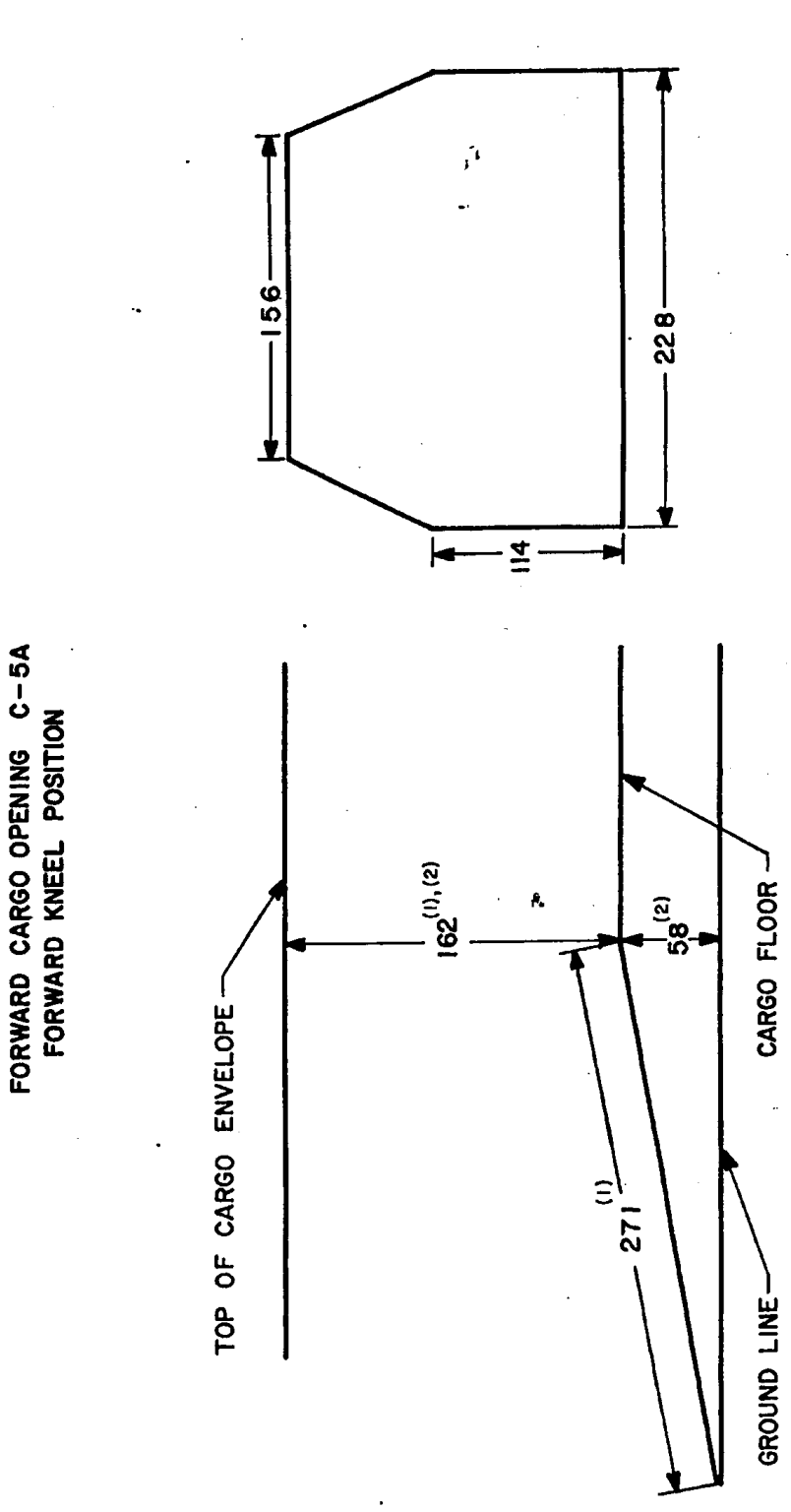
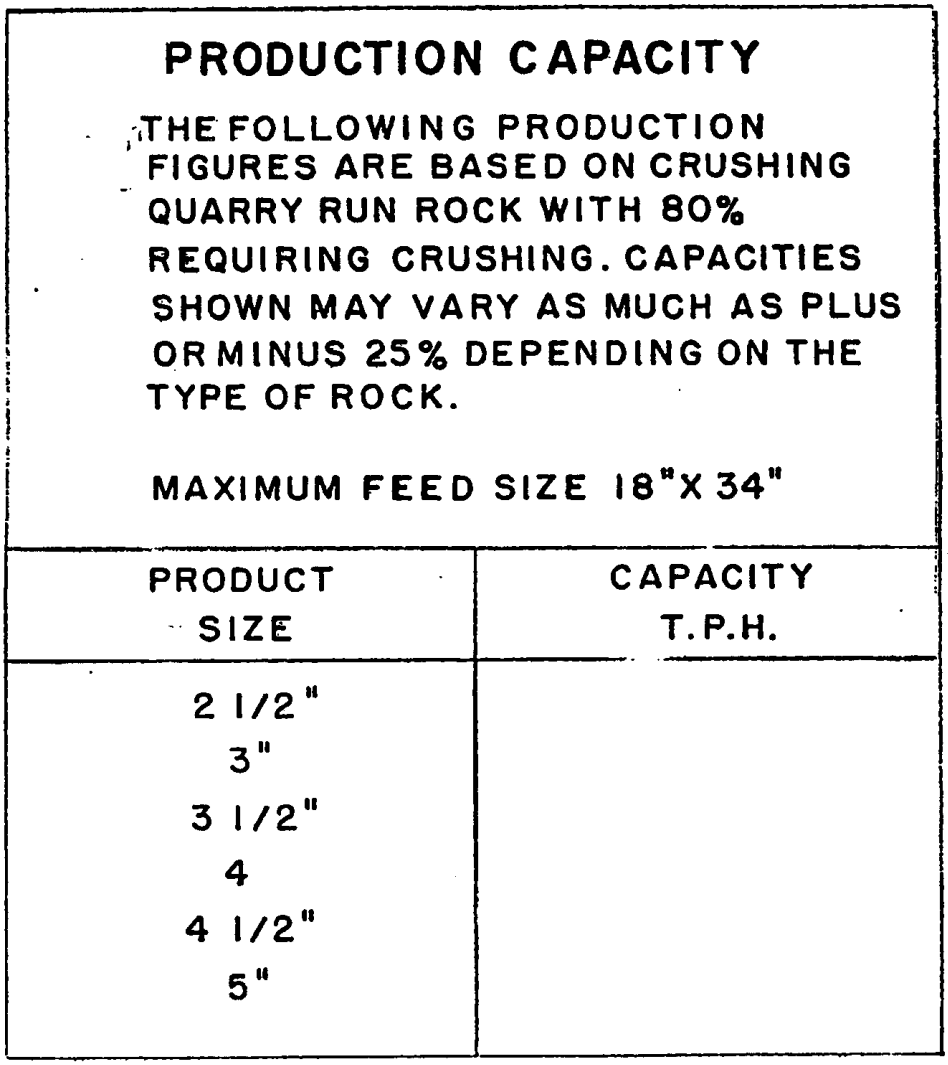


FIGURE 6. C-5A. cargo envelope.

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PRODUCTION CAPACITY

THE FOLLOWING PRODUCTION FIGURES ARE BASED ON CRUSHING QUARRY RUN ROCK WITH 80% REQUIRING CRUSHING. CAPACITIES SHOWN MAY VARY AS MUCH AS PLUS OR MINUS 25% DEPENDING ON THE TYPE OF ROCK.

MAXIMUM FEED SIZE 18"X 34"

PRODUCT SIZE	CAPACITY T.P.H.
2 1/2"	
3"	
3 1/2"	
4	
4 1/2"	
5"	

C13200 E6366

FIGURE 7. PRODUCTION CAPACITY PLATE.

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